Practical Digital System Design

Final Project

IoT Data Filtering

Group: 轟隆隆紅紅空空恐龍衝衝衝
Leader: 洪漢霖 B103012011
Member: 葉米亞 B103015019
Member: 賴佳華 B103015011

Abstract of Design and Implementation

Has RTL(pre-synthesis) verification passed?	Yes
Has gate-level(post-synthesis) verification passed?	Yes
Clock period (ns)	4
Area (um^2)	44563.539783
Power (mW)	3.3761 mW
Simulation time of the 7 functions (ns) (Gate-level Simulation)	43112 in total
Have you referenced the design concepts or architectures of other groups?	DYBP: using K-map to derive the logic of exclude and extract function
Coding style check and enhancement?	No
Significant features	Using KS adder

Team Contributions and Job Assignment

Please include the signature of all team members.

Team Member 1 [洪漢霖]: (37.5%)

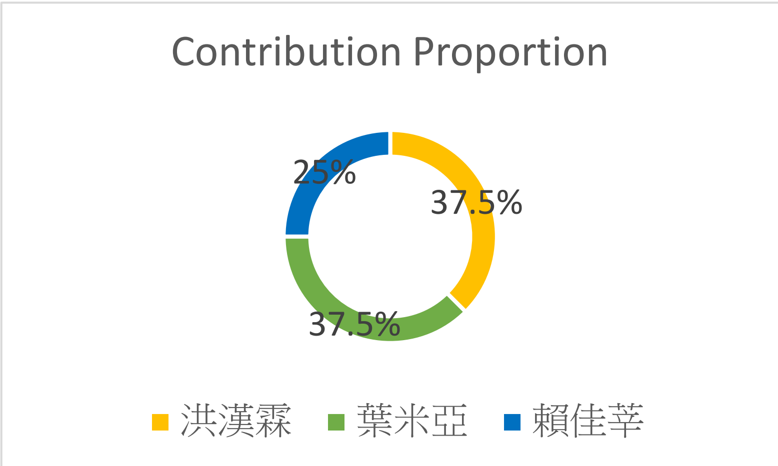
Leader, algorithm derivation, circuit design, presentation, circuit synthesis, synthesis script, and report writing.

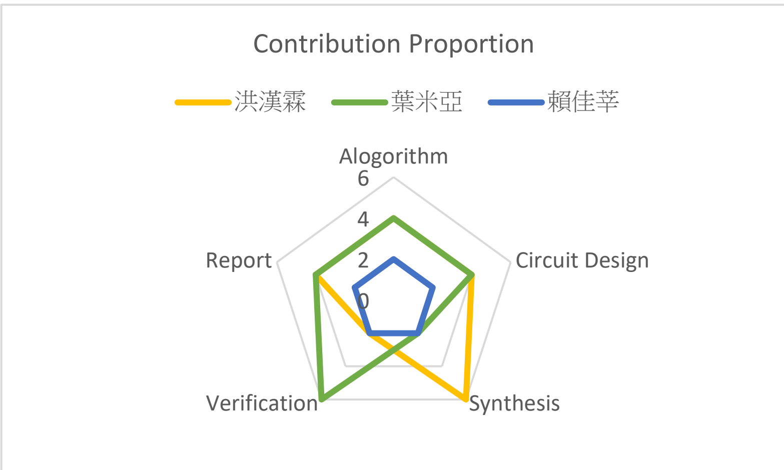
Team Member 2 [葉米亞]: (37.5%)

PowerPoint presentation creation, presentation, circuit design, waveform simulation, and circuit synthesis, and report writing.

Team Member 3 [賴佳莘]: (25%)

Algorithm derivation, circuit design, circuit synthesis





洪漢霖 葉米亞 賴佳莘

I. Algorithm Introduction and Explanation

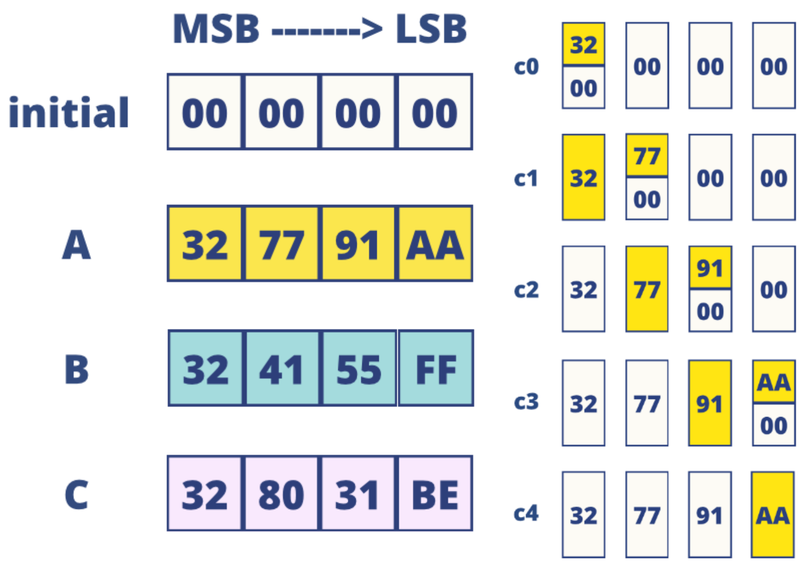
-function-

(1)MAX, MIN:

Since the difference between MAX and MIN lies only in the condition for determining the change. For the MAX case, change is set to 1 when the input is greater than the output, while for MIN, change is set to 0 when the input is less than the output.

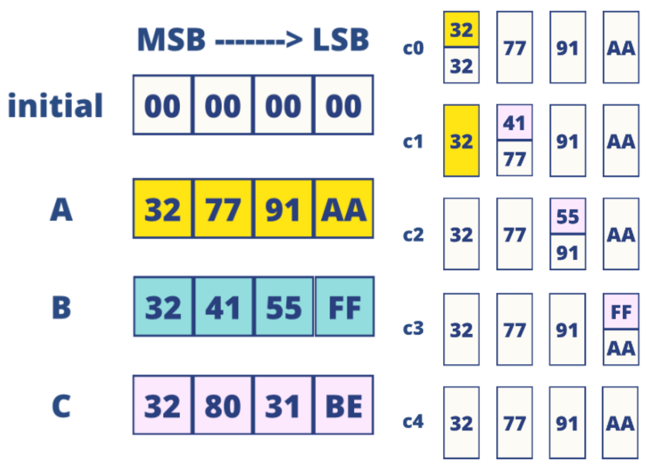
The following is an example explanation for MAX:
Assume we have the following input patterns(A,B,C,D)
When reset=1, the output_reg would has initial value of 128'b0.

Round 1-Pattern 1(A):



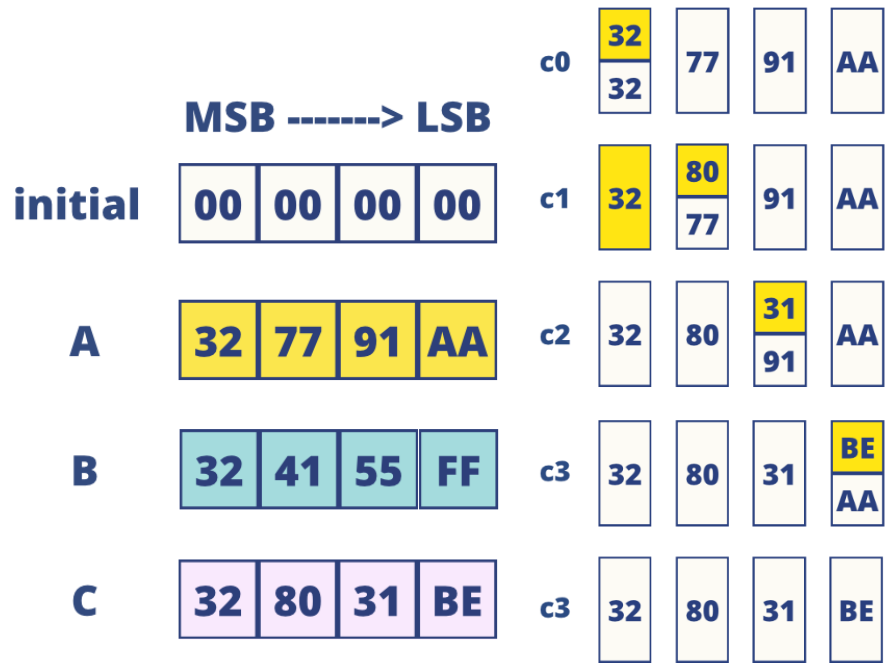
At cycle 0, the MSB of the input pattern A (32) is bigger than the MSB of the output_reg(0), the change would be 1 and lock its value until this pattern is done. Therefore, output_reg will keep updating the value from input pattern A until the pattern is done(pd=1).

Round 1-Pattern 2(B):



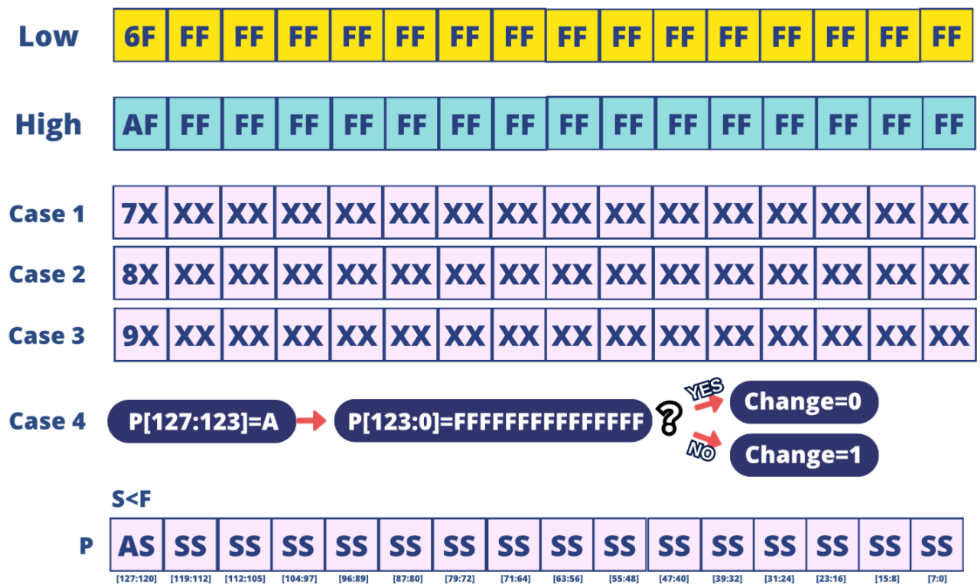
At cycle 0, the MSB of the input pattern B (32) is equal to the MSB of the output_reg(32), the change would be 0 .
At cycle 1, the next value of B is 41 and is less than 77, therefore, the change would be 0 and lock its value until this pattern is done. output_reg will keep remaining its original value until the pattern is done.

Round 1-Pattern 3(C):



At cycle 0, the MSB of the input pattern C (32) is equal to the MSB of the output_reg(32), the change would be 0 .
At cycle 1, the next value of C is 80 and is bigger than 77, therefore, the change would be 1 and lock its value until this pattern is done. Therefore, output_reg will keep updating the value from input pattern C until the pattern is done(pd=1).

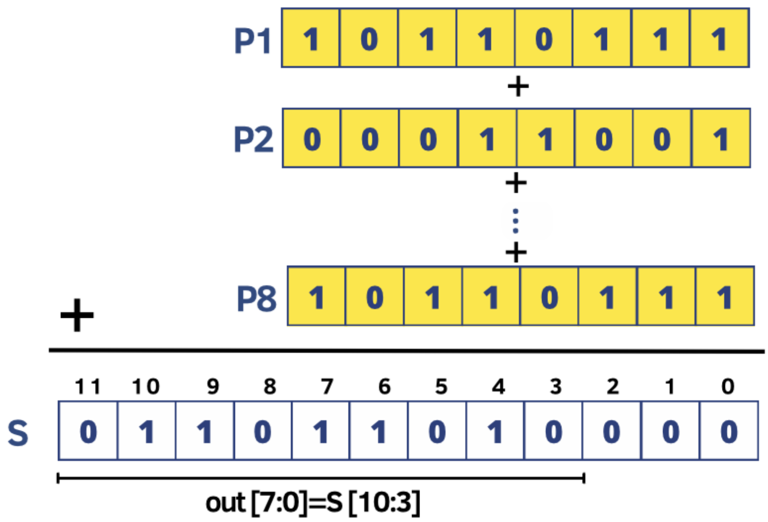
(2)EXTRACT, EXCLUDE:



Because the upper and lower bounds for 'extract' are fixed values (for EXTRACT: low = 128'h 6FFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF high = 128'h AFFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF_FFFF), the MSB of the input pattern can be used to determine if it falls between these two values. Cases 1 to 3 check if P[128:123] is 7, 8, or 9. In Case 4, if P[128:123] is A, it checks whether the subsequent bits are all F. If they are, it means it's not within this range, so change = 0. If they are not all F, it means it's within this range, so change = 1. Because the algorithms for EXTRACT and EXCLUDE are similar in mechanism, the following is an example using EXTRACT.
In this part, many groups turn to use the K-map to derive the logic of the input is valid or not. We choose to cite DYBP's concepts, since the concept is explained clearly and also naïve and intuitive to implement.

(3)PEAKMAX, PEAKMIN:
The comparison condition of PEAKMAX is similar to that of MAX(Similarly, PEAKMIN shares similarities with MIN) , with the difference lying in the evaluation occurring when the pattern counter is 0 and the cycle number is 15. In other words, this evaluation takes place after each round. During this check, the system determines whether the current input is greater than the existing output. If the current input surpasses the output value, the algorithm updates by writing the new value; otherwise, it retains the current value.

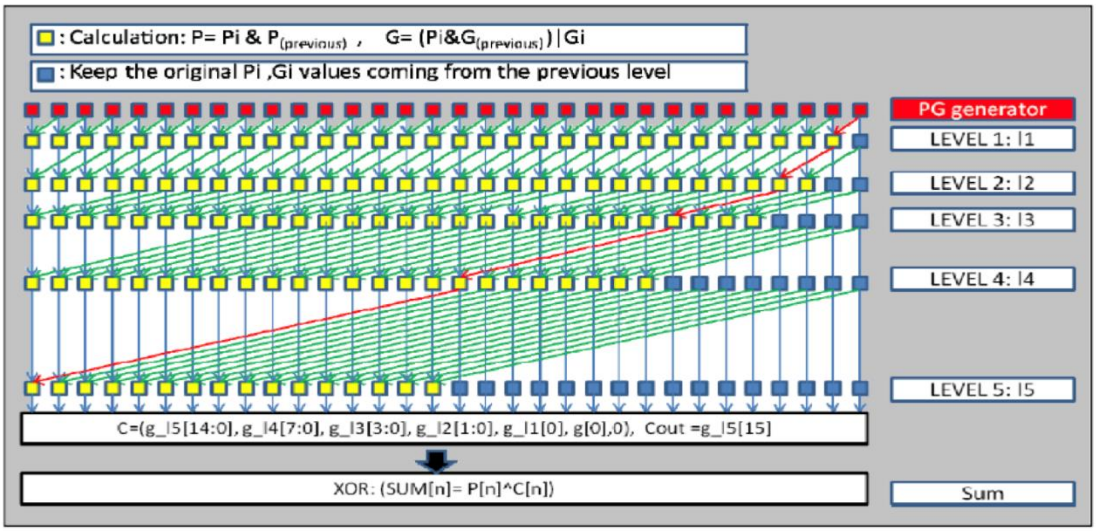
(4)AVE:



At the beginning of each pattern input, the new sum value is passed to the adder module for computing the updated sum. At the end of each round, the output value consists of the 128 most significant bits of the 131-bit sum, excluding bits [130:3].

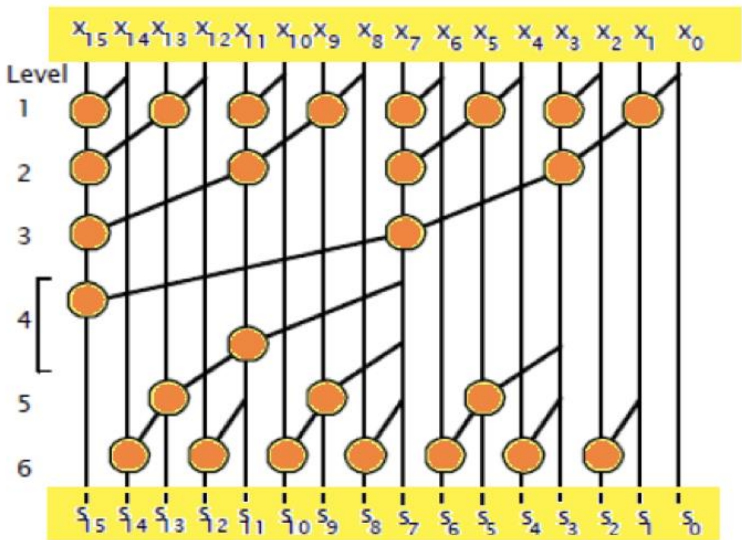
-Adder-

(1)Kogge-Stone Adder:



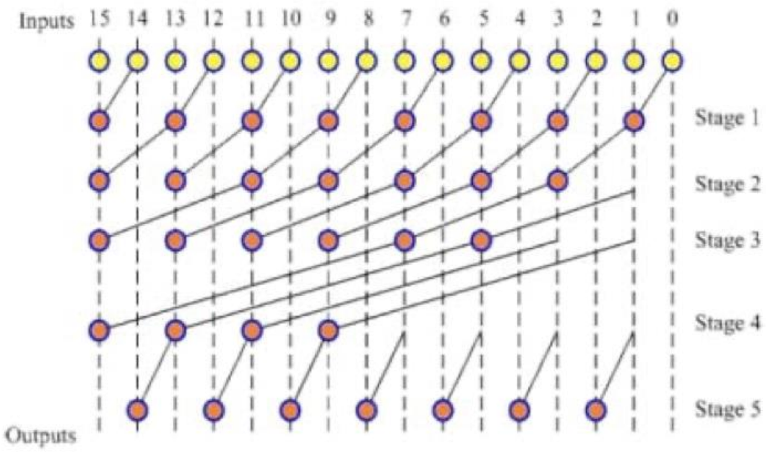
The Kogge Stone Adder is a high-performance adder widely utilized in various applications. Its fundamental concept is akin to the conventional carry lookahead adder, with a distinctive feature in the second stage known as the parallel carry prefix chain. In the initial level (L=1), the generation and propagation of 2 bits are computed concurrently. Progressing to the second level (L=2), the generation and propagation of 4 bits are determined using the results from the first level. This enables the availability of the actual carry-out value for the 4th bit while computations in level 2 are ongoing. The same approach continues for subsequent levels (L=3, 4, 5) to calculate carry-out values for the 8th, 16th, and 32nd bits. All carries for each bit are computed in parallel. In Figure 1, red boxes denote propagate (P) and generate (G) generators for each bit of the two inputs. Yellow boxes contain propagate and generate blocks, while blue boxes retain the original generate values transmitted from the preceding level. The delay at each level is determined by the runtime of a single yellow box.

(2)Han-Carlson Adder:



The Han Carlson Adder employs a parallel prefix tree design, contributing to complexity reduction in comparison to the Brent Kung Adder. It represents a hybrid design incorporating stages from both the Brent Kung and Kogge Stone adders. This design performs carry-merge operations exclusively on even bits. Generate and propagate signals of odd bits are transmitted down the prefix tree and recombine with even bits' carry signals at the end to produce true carry bits. The reduced complexity comes at the cost of adding an extra stage to its carry-merge path. Figure 3 illustrates the methodology of a 16-bit Han Carlson Adder.

(3)Brent-Kung Adder:



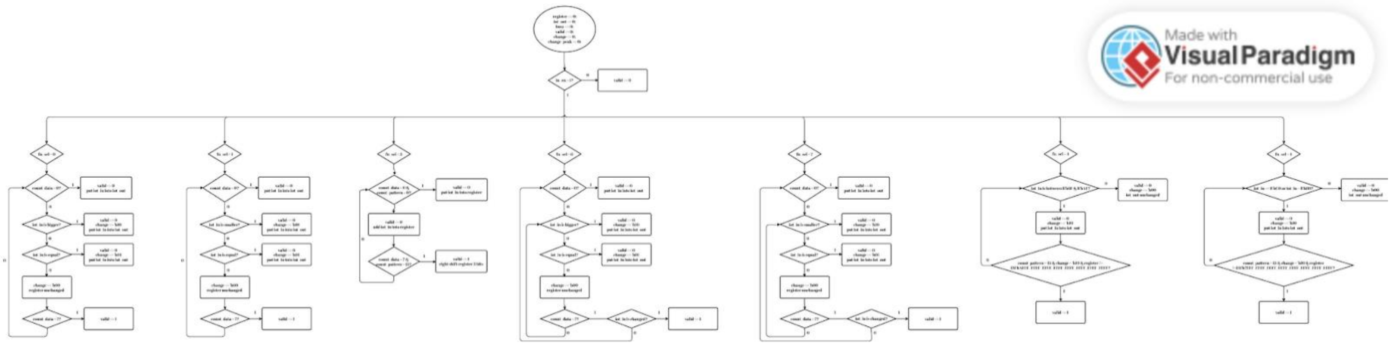
The Brent Kung Adder is a parallel prefix form of the carry lookahead adder. It aims to achieve an O(log2(n)) gate-level depth as the size of input operands increases, reducing the delay compared to traditional prefix adders such as the Kogge Stone adder. The schematic in Figure 2 illustrates an approach that avoids using a carry chain to calculate the output, minimizing delay without compromising power performance. It occupies less area and experiences reduced wiring congestion.

II. Architecture Introduction and Explanation

(Include the circuit architecture diagram)

Method:

-Method I-



-Method II-

(1) Adder Comparison:

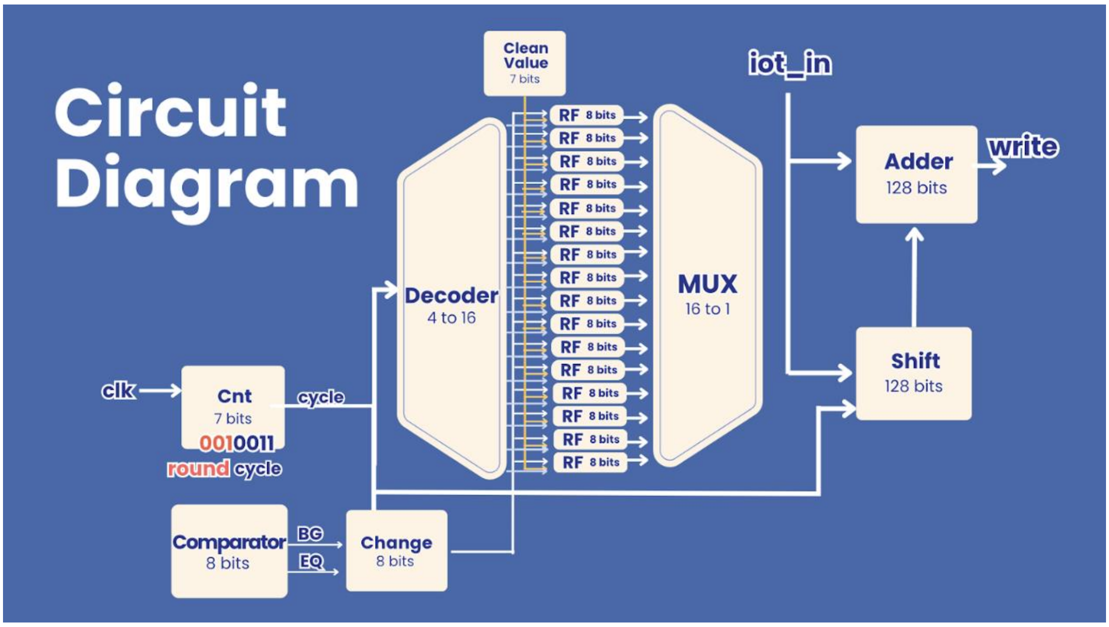
We implemented three different adders using three distinct algorithms, namely the Kogge-Stone adder, Han-Carlson adder, and Brent-Kung adder.
And the following is their comparison:

COMPARISON OF ADDERS

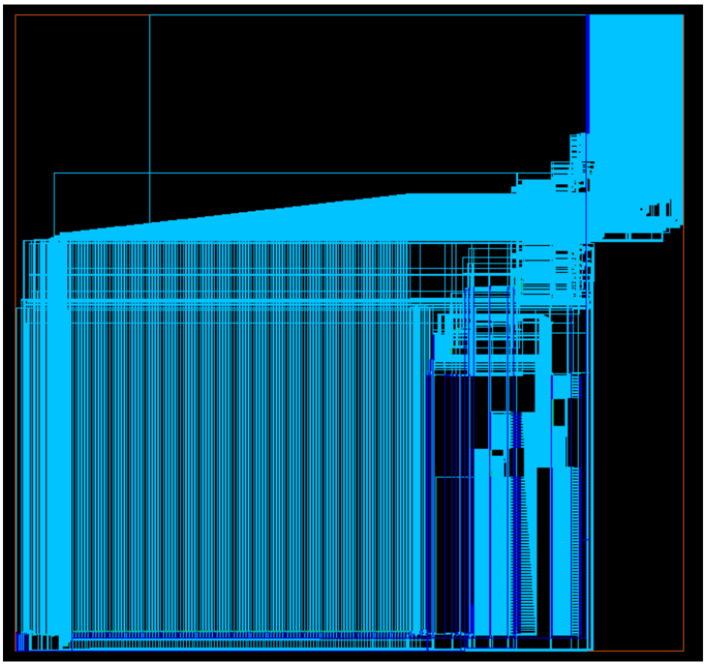
	Kogge-Stone Adder	Han-Carlson Adder	Brent-Kung Adder
Pros	• Excellent Parallelism • Low Latency	• Low Power • Consumption	• High Parallelism • Low Area Complexity
Cons	• Complex Area • Higher Power Consumption	Higher Latency	• Potential Higher Latency
Area	11196.1(um²)	7930.2(um²)	7685.8(um²)
Delay	3.23(ns)	6.85(ns)	6.85(ns)

Although the area of KS adder is bigger than HC adder and BK adder for about 41%, however, the delay is 53% less than HC adder and BK adder.
And since the score is Area X Delay, we think that using KS adder would be the best choice.

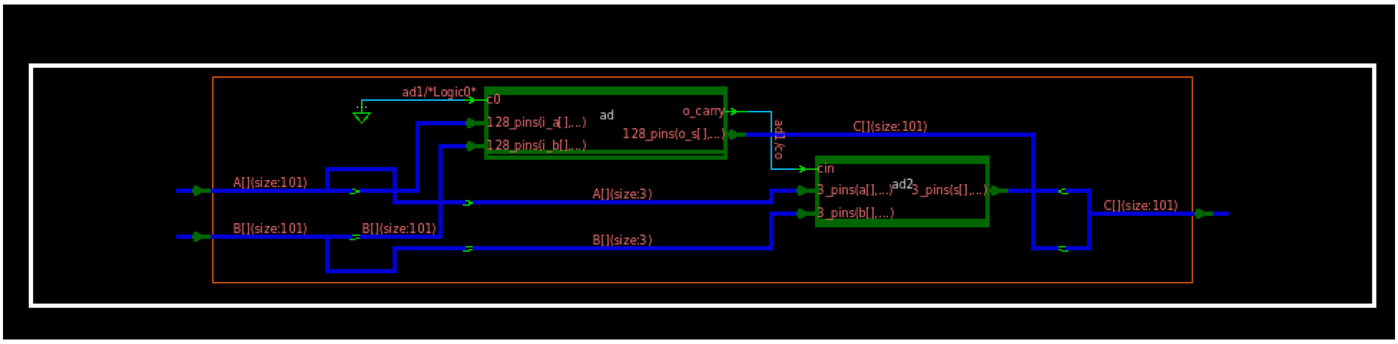
Circuit diagram-



For our main circuit, it acts like a Register file. In the beginning, we tried to use decoder and mux to control the write enable.
Decoder: counter change in to 16bit onehot code of write enable.
RF128: if write enable==1, value comes from outside, or it'll remain the old one.
MUX: using counter to decide which old value should be compared with current input.
Adder: add ALL data from RF and the input reg(act like buffers), store back into RF.
Shift: in the beginning, we tried to use shifter to add once there's a new input, finally, we use the input reg, though the larger area is consumed, however, there's lower delay.
Change: we use det(determined), udf(undefined) to control the state that new input pattern should be written or not.



Since we do not tear apart the module, the synthesis result is quite hard to observe.

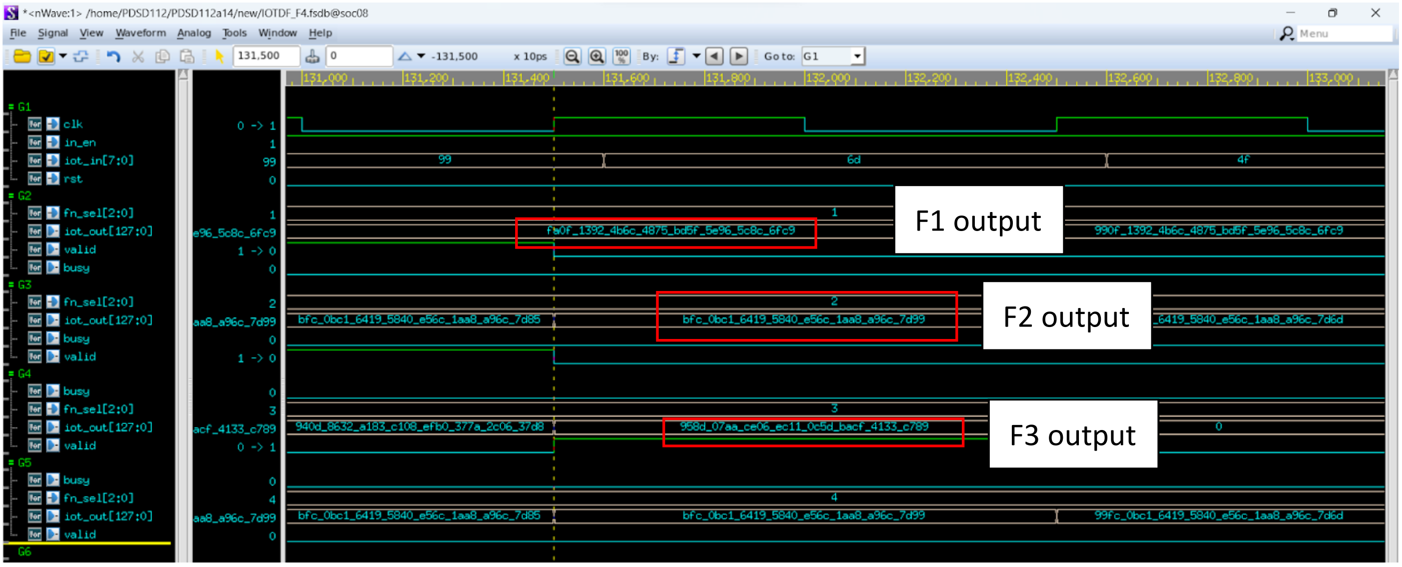


However, our adder is composed of 128bitKSA and 3bit CLA, so the structure is easy to observe.

III. Verification Process Explanation

1. Original TestBench

-Round 0-




```
1A_A9_92_A8_74_2B_E4_86_B7_89_00_B6_8F_C1_38_1D      //Round_00  P_00: 3.544040e+37
D1_99_D7_A0_25_9F_50_AA_B8_CA_1A_89_78_5D_99_5B      //Round_00  P_01: 2.786074e+38
FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9      //Round_00  P_02: 3.323853e+38
C0_E4_42_33_61_AD_10_37_B0_C0_EA_36_5F_38_2C_4E      //Round_00  P_03: 2.563970e+38
E8_F4_B0_68_EC_2D_0F_38_42_A7_FF_69_62_41_31_BB      //Round_00  P_04: 3.096514e+38
9B_2E_6E_48_31_E3_94_E3_02_8D_2B_84_41_BC_30_A5      //Round_00  P_05: 2.062714e+38
75_12_52_D5_A7_28_D6_4D_59_D9_2C_D6_F8_50_EE_D6      //Round_00  P_06: 1.556148e+38
0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85      //Round_00  P_07: 1.593021e+37
```

(1)MAX :since the max value is :

```
FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9      //Round_00
```

=> correct!

(2)MIN:since the min value is :

```
0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85        //Round_00
```

=> correct!

(3)AVE: since the average value is :

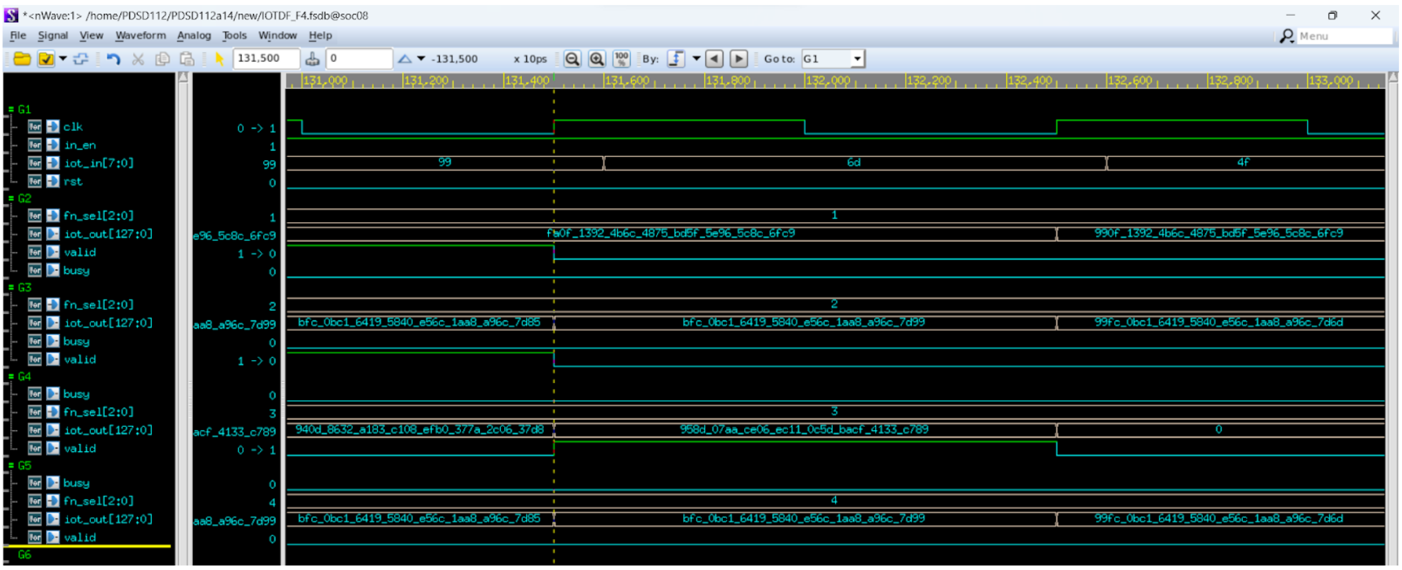
```
95_8D_07_AA_CE_06_EC_11_0C_5D_BA_CF_41_33_C7_89_       //Round_00
```

=> correct!

(4)Extract: since the between low and high show in the following

```
9B_2E_6E_48_31_E3_94_E3_02_8D_2B_84_41_BC_30_A5        //Round_00
75_12_52_D5_A7_28_D6_4D_59_D9_2C_D6_F8_50_EE_D6        //Round_00
```

=> correct!



(5)Exclude: since the values below are not between low and high

```
1A_A9_92_A8_74_2B_E4_86_B7_89_00_B6_8F_C1_38_1D      //Round_00
D1_99_D7_A0_25_9F_50_AA_B8_CA_1A_89_78_5D_99_5B      //Round_00
FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9      //Round_00
C0_E4_42_33_61_AD_10_37_B0_C0_EA_36_5F_38_2C_4E      //Round_00
E8_F4_B0_68_EC_2D_0F_38_42_A7_FF_69_62_41_31_BB      //Round_00
75_12_52_D5_A7_28_D6_4D_59_D9_2C_D6_F8_50_EE_D6      //Round_00
0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85      //Round_00
```

=> correct!

(6)PeakMax:

Since this is round 0, the value must be peakmax

```
FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9      //Round_00
```

=> correct!

(7)PeakMin:

Since this is round 0, the value must be peakmin

```
0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85      //Round_00
```

=> correct!

Round 1:

```
99_6D_4F_DD_31_C2_59_E5_BE_93_9B_7B_81_BF_CB_61      //Round_01   P_00: 2.039395e+38
C8_C2_6D_6A_08_E4_0A_1E_4B_C4_69_DA_32_5A_2F_9E      //Round_01   P_01: 2.668551e+38
D1_EC_9D_BE_CD_3F_64_AA_2F_8A_41_7B_1A_72_3A_9A      //Round_01   P_02: 2.790372e+38
39_DF_3E_11_50_98_DD_53_29_FB_33_45_E3_08_D9_89      //Round_01   P_03: 7.692514e+37
BB_F6_D8_F5_BF_13_06_53_13_1B_30_BB_5C_BA_7E_BA      //Round_01   P_04: 2.498473e+38
AD_4B_E1_AF_97_44_2B_D3_6F_4D_E2_F8_74_9F_FC_60      //Round_01   P_05: 2.303504e+38
F8_B5_ED_42_4B_D8_D4_8C_45_61_A1_60_10_20_2C_1F      //Round_01   P_06: 3.305932e+38
69_E6_90_BC_D0_AF_03_A5_3C_79_9F_3D_D9_37_F3_5D      //Round_01   P_07: 1.407661e+38
```

(1)MAX :since the max value is :

```
F8_B5_ED_42_4B_D8_D4_8C_45_61_A1_60_10_20_2C_1F        //Round_01
```

=> correct!

(2)MIN:since the min value is :

```
39_DF_3E_11_50_98_DD_53_29_FB_33_45_E3_08_D9_89        //Round_01
```

=> correct!

(3)AVE: since the average value is :

```
 A7_3B_5A_37_79_6B_B6_0B_2D_04_39_CD_0D_88_F5_37_        //Round_01
```

=> correct!

(4)Extract: since the between low and high show in the following

```
99_6D_4F_DD_31_C2_59_E5_BE_93_9B_7B_81_BF_CB_61        //Round_01
AD_4B_E1_AF_97_44_2B_D3_6F_4D_E2_F8_74_9F_FC_60        //Round_01
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
C8_C2_6D_6A_08_E4_0A_1E_4B_C4_69_DA_32_5A_2F_9E        //Round_01
D1_EC_9D_BE_CD_3F_64_AA_2F_8A_41_7B_1A_72_3A_9A        //Round_01
39_DF_3E_11_50_98_DD_53_29_FB_33_45_E3_08_D9_89        //Round_01
F8_B5_ED_42_4B_D8_D4_8C_45_61_A1_60_10_20_2C_1F        //Round_01
69_E6_90_BC_D0_AF_03_A5_3C_79_9F_3D_D9_37_F3_5D        //Round_01
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9 .All the patterns in round 1 aren’t bigger than peakmax, therefore the PeakMax doesn’t output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is 0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85 .All the patterns in round 1 aren’t smaller than peakmin, therefore the PeakMin doesn’t output anything.

=> correct!

-Round 2-

```
0D_DF_4D_7A_E4_7C_77_D8_25_38_A8_3E_47_4E_61_CF       //Round_02  P_00: 1.843942e+37
50_D4_C5_DF_96_5B_D4_98_78_52_2D_04_E9_07_72_E5       //Round_02  P_01: 1.074430e+38
50_09_81_52_D0_36_E6_54_BA_3A_A0_FB_0F_33_5C_58       //Round_02  P_02: 1.063876e+38
7E_E7_9C_8F_8E_3A_81_36_64_05_E0_CD_9D_11_24_04       //Round_02  P_03: 1.686853e+38
0E_A0_AF_3E_D0_C3_D6_A0_27_03_2D_90_90_00_D1_07       //Round_02  P_04: 1.944351e+37
18_92_15_22_09_65_91_43_24_C8_A5_63_AB_A2_2E_D9       //Round_02  P_05: 3.265998e+37
3D_BC_B6_3C_D9_D2_C6_F9_E7_04_E2_A7_55_CD_D5_F0       //Round_02  P_06: 8.206276e+37
C8_82_4C_5D_1D_86_50_29_5A_E7_40_FE_09_64_68_C3       //Round_02  P_07: 2.665221e+38
```

(1)MAX :since the max value is :

```
C8_82_4C_5D_1D_86_50_29_5A_E7_40_FE_09_64_68_C3       //Round_02
```

=> correct!

(2)MIN:since the min value is :

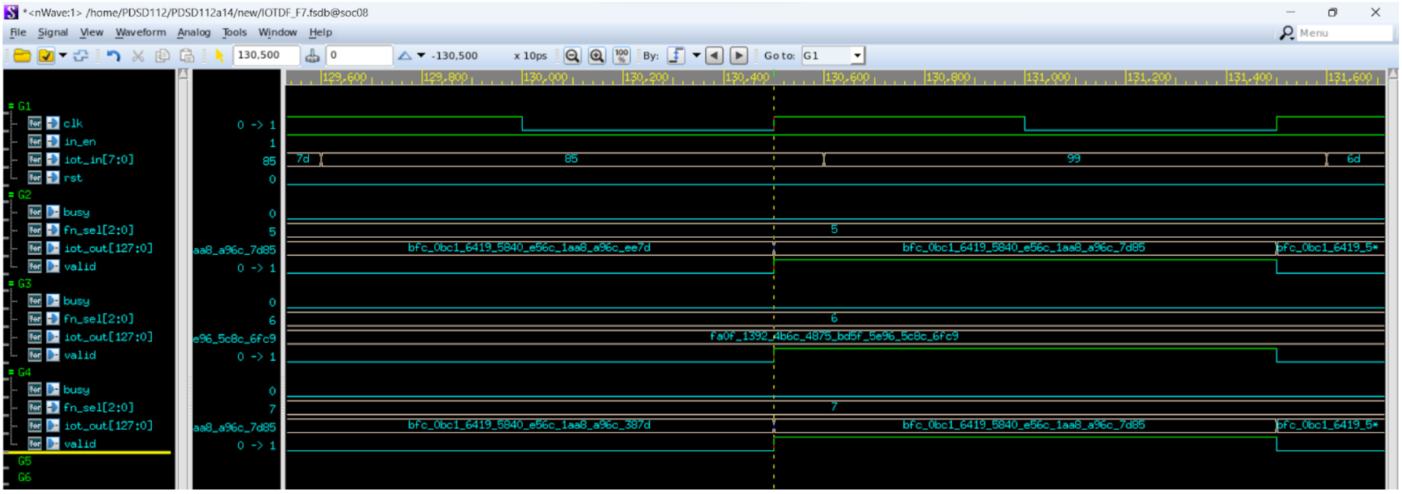
```
0D_DF_4D_7A_E4_7C_77_D8_25_38_A8_3E_47_4E_61_CF       //Round_02
```

=> correct!

(3)AVE: since the average value is :

```
4B_62_DF_06_F5_59_86_60_49_30_69_B4_AE_ED_F2_74_       //Round_02
```

=> correct!



(4)Extract: since the between low and high show in the following

```
7E_E7_9C_8F_8E_3A_81_36_64_05_E0_CD_9D_11_24_04        //Round_02
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
0D_DF_4D_7A_E4_7C_77_D8_25_38_A8_3E_47_4E_61_CF        //Round_02
50_D4_C5_DF_96_5B_D4_98_78_52_2D_04_E9_07_72_E5        //Round_02
50_09_81_52_D0_36_E6_54_BA_3A_A0_FB_0F_33_5C_58        //Round_02
7E_E7_9C_8F_8E_3A_81_36_64_05_E0_CD_9D_11_24_04        //Round_02
0E_A0_AF_3E_D0_C3_D6_A0_27_03_2D_90_90_00_D1_07        //Round_02
18_92_15_22_09_65_91_43_24_C8_A5_63_AB_A2_2E_D9        //Round_02
3D_BC_B6_3C_D9_D2_C6_F9_E7_04_E2_A7_55_CD_D5_F0        //Round_02
C8_82_4C_5D_1D_86_50_29_5A_E7_40_FE_09_64_68_C3        //Round_02
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9 .All the patterns in round 2 aren't bigger than peakmax, therefore the PeakMax doesn't output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is 0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85 .All the patterns in round 2 aren't smaller than peakmin, therefore the PeakMin doesn't output anything.

=> correct!

Round 3:

```
DE_DD_3B_B0_26_1A_1B_CB_28_32_96_B6_49_43_6B_F7      //Round_03  P_00: 2.962373e+38
9F_23_51_09_BA_0C_B1_EC_DF_9B_7C_1A_EE_48_88_7E      //Round_03  P_01: 2.115306e+38
55_A8_48_B0_33_F0_BD_77_CD_85_D6_72_CC_B6_6E_8E      //Round_03  P_02: 1.138582e+38
45_97_40_42_BC_9D_65_9D_BB_B4_0D_B1_54_31_00_A6      //Round_03  P_03: 9.250207e+37
08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9      //Round_03  P_04: 1.131466e+37
DE_3E_AB_8B_35_7A_0D_F8_66_C2_8D_EB_91_3F_41_E7      //Round_03  P_05: 2.954140e+38
3F_F9_47_75_BA_FB_5F_A7_DC_81_6D_A1_04_EF_97_06      //Round_03  P_06: 8.503570e+37
FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93      //Round_03  P_07: 3.361817e+38
```

(1)MAX :since the max value is :

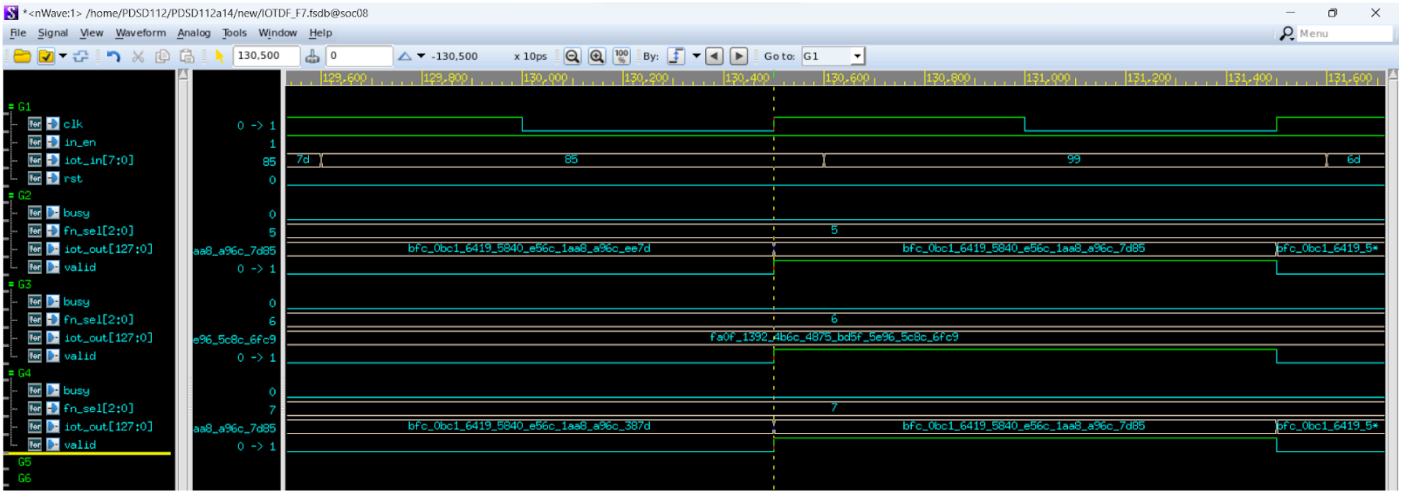
```
FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93      //Round_03
```

=> correct!

(2)MIN: since the min value is :

```
08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9        //Round_03
```

=> correct!



(3)AVE: since the average value is :

```
87_9C_AC_F6_27_97_42_93_90_0D_6E_F8_C2_F8_E8_64_        //Round_03
```

=> correct!

(4)Extract: since the between low and high show in the following

```
9F_23_51_09_BA_0C_B1_EC_DF_9B_7C_1A_EE_48_88_7E        //Round_03
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
DE_DD_3B_B0_26_1A_1B_CB_28_32_96_B6_49_43_6B_F7        //Round_03
55_A8_48_B0_33_F0_BD_77_CD_85_D6_72_CC_B6_6E_8E        //Round_03
45_97_40_42_BC_9D_65_9D_BB_B4_0D_B1_54_31_00_A6        //Round_03
08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9        //Round_03
DE_3E_AB_8B_35_7A_0D_F8_66_C2_8D_EB_91_3F_41_E7        //Round_03
3F_F9_47_75_BA_FB_5F_A7_DC_81_6D_A1_04_EF_97_06        //Round_03
FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93        //Round_03
```

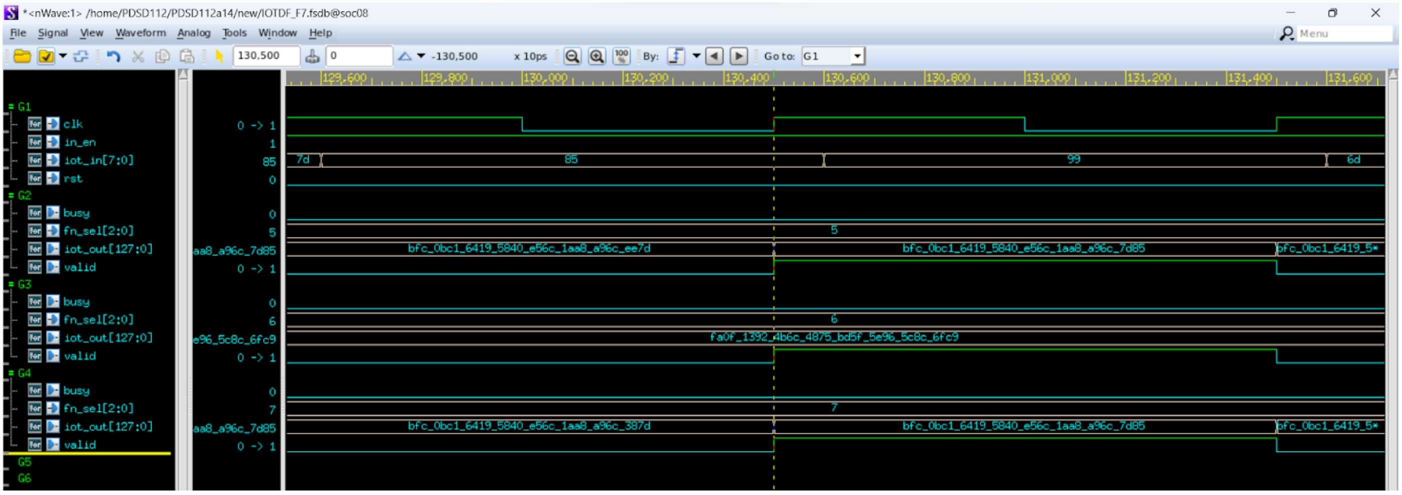
=> correct!

(6)PeakMax:

Since the peakmax value now is FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9 .
The patterns of FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93 in round 3 is bigger than peakmax, therefore the PeakMax output the following pattern

```
FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93        //Round_03
```

=> correct!



(7)PeakMin:

Since the peakmin value now is 0B_FC_0B_C1_64_19_58_40_E5_6C_1A_A8_A9_6C_7D_85 .
The patterns of 08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9 in round 3 is smaller than peakmin, therefore the PeakMin output the following pattern

```
08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9        //Round_03
```

=> correct!

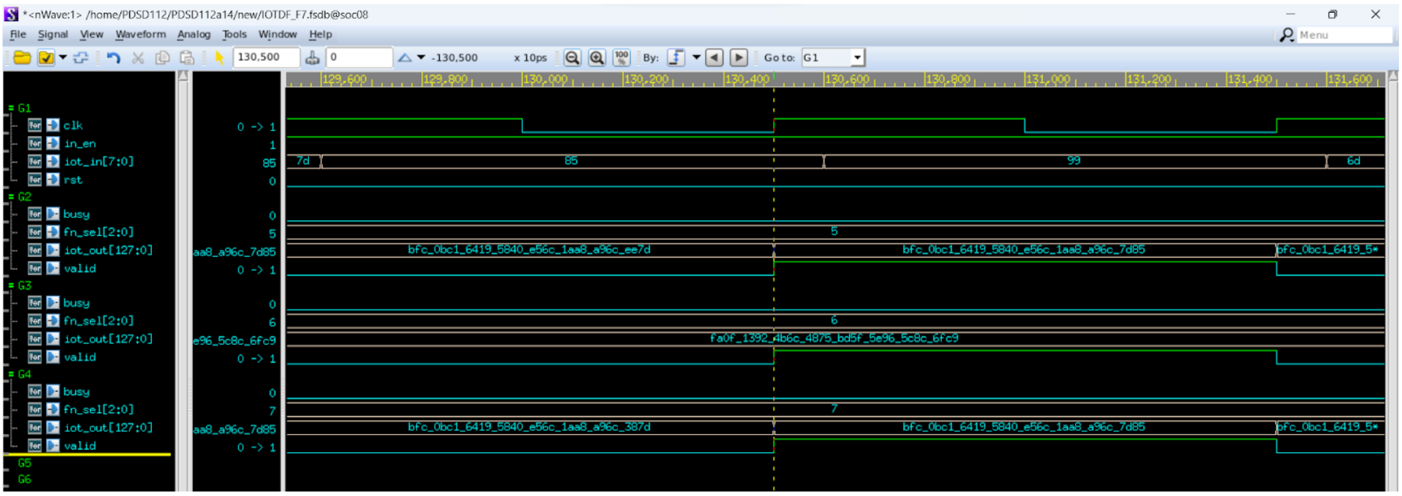
Round 4:

```
20_A6_91_41_83_3B_48_6A_EC_5C_65_6D_07_E1_8C_BA      //Round_04  P_00: 4.340016e+37
88_1C_0D_64_F5_1D_EA_26_D2_EC_5B_4C_C8_FE_C7_26      //Round_04  P_01: 1.809207e+38
60_83_D2_F6_2F_21_A8_7E_AF_E6_A8_E3_FB_F2_8F_0C      //Round_04  P_02: 1.282904e+38
C8_7B_09_84_33_BD_6C_D9_78_8D_64_04_B0_7A_4D_88      //Round_04  P_03: 2.664844e+38
EA_11_44_1E_A8_23_A0_AD_E3_85_2D_25_C7_1F_75_0A      //Round_04  P_04: 3.111290e+38
5C_63_4D_EA_BD_2F_99_72_FB_2D_0D_F5_99_10_EB_7F      //Round_04  P_05: 1.228046e+38
E9_CE_A9_74_5F_65_FB_7F_08_9B_3F_B6_48_62_E2_35      //Round_04  P_06: 3.107832e+38
1E_C9_A0_E7_F9_56_C3_27_D3_BA_E6_4E_BF_8E_E1_79      //Round_04  P_07: 4.092376e+37
```

(1)MAX :since the max value is :

```
EA_11_44_1E_A8_23_A0_AD_E3_85_2D_25_C7_1F_75_0A        //Round_04
```

=> correct!



(2)MIN:since the min value is :

```
1E_C9_A0_E7_F9_56_C3_27_D3_BA_E6_4E_BF_8E_E1_79        //Round_04
```

=> correct!

(3)AVE: since the average value is :

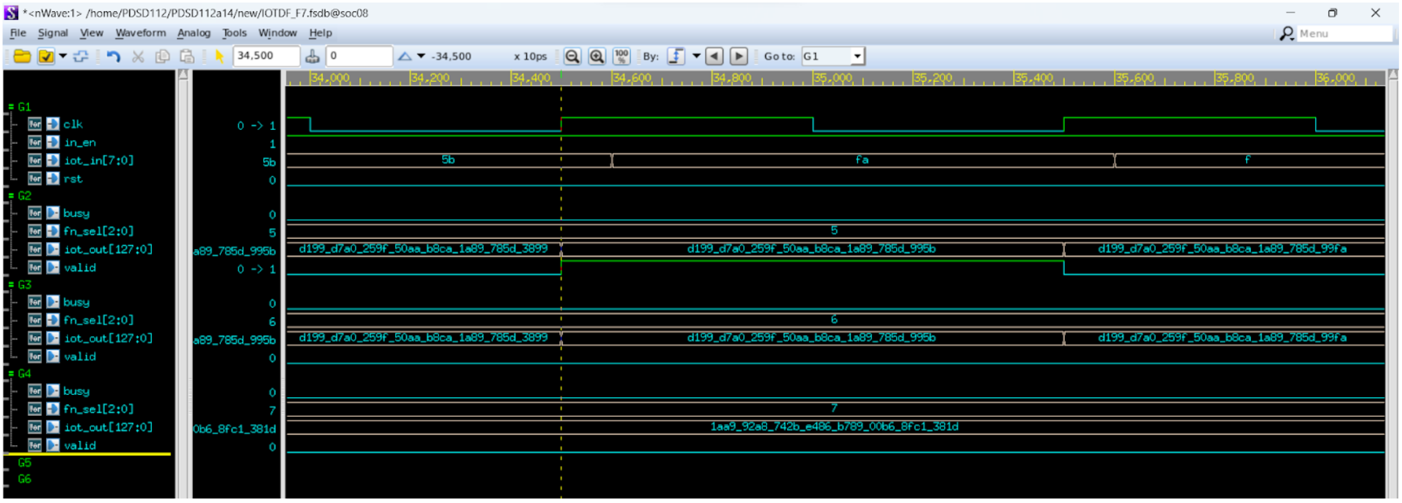
```
84_19_CA_F0_D3_29_08_16_34_58_A5_D8_5C_AD_EA_95_        //Round_04
```

=> correct!

(4)Extract: since the between low and high show in the following

```
88_1C_0D_64_F5_1D_EA_26_D2_EC_5B_4C_C8_FE_C7_26        //Round_04
```

=> correct!



(5)Exclude: since the values below are not between low and high

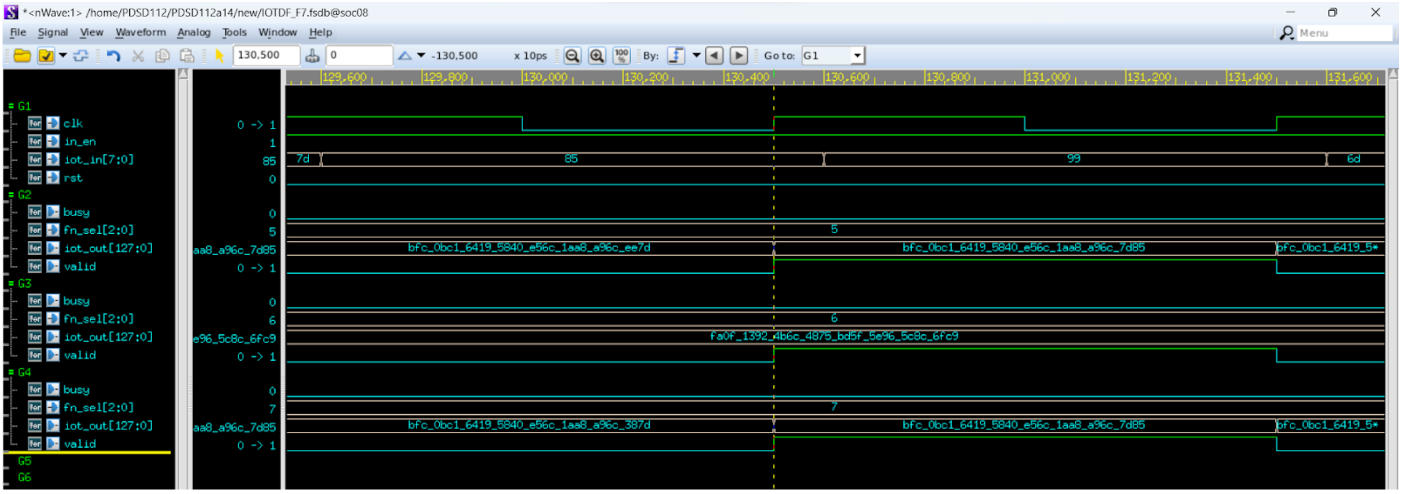
```
20_A6_91_41_83_3B_48_6A_EC_5C_65_6D_07_E1_8C_BA        //Round_04
60_83_D2_F6_2F_21_A8_7E_AF_E6_A8_E3_FB_F2_8F_0C        //Round_04
C8_7B_09_84_33_BD_6C_D9_78_8D_64_04_B0_7A_4D_88        //Round_04
EA_11_44_1E_A8_23_A0_AD_E3_85_2D_25_C7_1F_75_0A        //Round_04
5C_63_4D_EA_BD_2F_99_72_FB_2D_0D_F5_99_10_EB_7F        //Round_04
E9_CE_A9_74_5F_65_FB_7F_08_9B_3F_B6_48_62_E2_35        //Round_04
1E_C9_A0_E7_F9_56_C3_27_D3_BA_E6_4E_BF_8E_E1_79        //Round_04
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_939 .All the patterns in round 4 aren't bigger than peakmax, therefore the PeakMax doesn't output anything.

=> correct!



(7)PeakMin:

Since the peakmin value now is08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9 .All the patterns in round 4 aren’t smaller than peakmin, therefore the PeakMin doesn’t output anything.

=> correct!

Round 5:

```
08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7      //Round_05  P_00: 1.079329e+37
E7_69_46_C8_54_1A_E1_34_72_79_7B_CE_71_EF_C0_0C      //Round_05  P_01: 3.075983e+38
EE_CC_FE_0E_A6_2B_02_AD_92_C0_25_E8_6B_D3_03_DB      //Round_05  P_02: 3.174206e+38
97_40_4C_36_0A_A4_92_1F_CC_2E_82_A4_26_2E_FE_0A      //Round_05  P_03: 2.010473e+38
0E_96_80_C6_22_C9_F8_98_A1_05_FE_A0_8A_64_74_A3      //Round_05  P_04: 1.939065e+37
E1_09_21_B2_3D_5F_88_C5_CA_21_C3_1E_32_E9_85_81      //Round_05  P_05: 2.991237e+38
D9_AE_E5_E8_DB_4E_10_94_0F_DC_F7_EB_BF_88_16_93      //Round_05  P_06: 2.893506e+38
91_D3_3C_9B_EA_E5_90_AF_63_CB_EE_9F_B7_73_C8_E1      //Round_05  P_07: 1.938349e+38
```

(1)MAX :since the max value is :

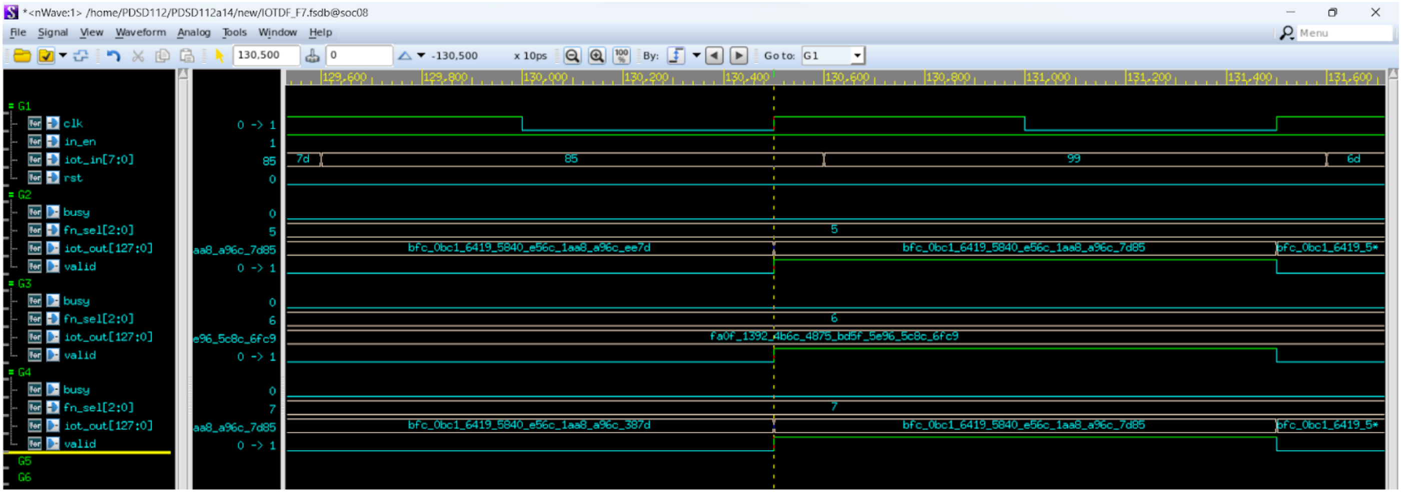
```
EE_CC_FE_0E_A6_2B_02_AD_92_C0_25_E8_6B_D3_03_DB      //Round_05
```

=> correct!

(2)MIN:since the min value is :

```
08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7      //Round_05
```

=> correct!



(3)AVE: since the average value is :

```
9A_16_E1_8A_86_97_AA_2A_CB_23_39_0F_1D_45_12_86_      //Round_05
```

=> correct!

(4)Extract: since the between low and high show in the following

```
97_40_4C_36_0A_A4_92_1F_CC_2E_82_A4_26_2E_FE_0A      //Round_05
91_D3_3C_9B_EA_E5_90_AF_63_CB_EE_9F_B7_73_C8_E1      //Round_05
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7      //Round_05
E7_69_46_C8_54_1A_E1_34_72_79_7B_CE_71_EF_C0_0C      //Round_05
EE_CC_FE_0E_A6_2B_02_AD_92_C0_25_E8_6B_D3_03_DB      //Round_05
0E_96_80_C6_22_C9_F8_98_A1_05_FE_A0_8A_64_74_A3      //Round_05
E1_09_21_B2_3D_5F_88_C5_CA_21_C3_1E_32_E9_85_81      //Round_05
D9_AE_E5_E8_DB_4E_10_94_0F_DC_F7_EB_BF_88_16_93      //Round_05
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93 .
All the patterns in round 5 aren't bigger than peakmax, therefore the PeakMax doesn't output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is08_83_1F_DB_DD_FD_58_69_7D_5A_BD_55_BC_94_AA_F9 .
The patterns of 08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7 in round 5 is smaller than peakmin, therefore the PeakMin output the following pattern

```
08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7       //Round_05
```

=> correct!

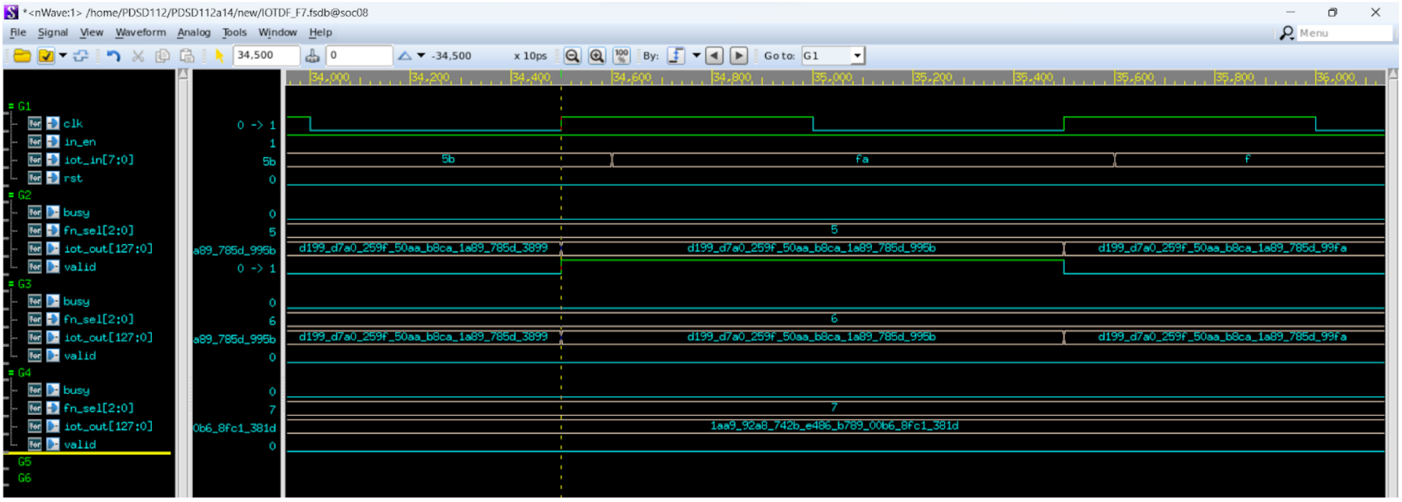
Round 6:

```
75_B3_11_8B_01_B9_3A_6D_6B_30_3C_41_B4_49_B7_2F      //Round_06   P_00: 1.564495e+38
11_5A_5D_C0_3F_68_70_46_AA_2B_18_F3_BC_9F_8B_33      //Round_06   P_01: 2.306608e+37
62_7D_FD_E0_9C_2F_19_FA_A2_AC_68_0B_A8_2E_BE_21      //Round_06   P_02: 1.309185e+38
BD_48_36_E4_FC_0D_20_F7_E2_66_21_05_E4_ED_AF_D2      //Round_06   P_03: 2.515990e+38
85_44_49_DB_FA_F1_CD_BA_9C_EF_C1_16_7D_B6_9C_BC      //Round_06   P_04: 1.771419e+38
7F_C3_81_AE_9E_6B_41_B9_9D_79_59_84_57_D5_15_E4      //Round_06   P_05: 1.698271e+38
D4_95_A1_68_F2_98_7F_A5_E0_66_73_E0_F6_7F_60_28      //Round_06   P_06: 2.825733e+38
04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C      //Round_06   P_07: 6.030286e+36
```

(1)MAX :since the max value is :

```
D4_95_A1_68_F2_98_7F_A5_E0_66_73_E0_F6_7F_60_28       //Round_06
```

=> correct!



(2)MIN:since the min value is :

```
04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C      //Round_06
```

=> correct!

(3)AVE: since the average value is :

```
70_9F_4E_A2_18_9A_DF_3A_CF_2D_B4_05_1D_40_E5_51_     //Round_06
```

=> correct!

(4)Extract: since the between low and high show in the following

```
75_B3_11_8B_01_B9_3A_6D_6B_30_3C_41_B4_49_B7_2F      //Round_06
85_44_49_DB_FA_F1_CD_BA_9C_EF_C1_16_7D_B6_9C_BC      //Round_06
7F_C3_81_AE_9E_6B_41_B9_9D_79_59_84_57_D5_15_E4      //Round_06
```

=> correct!

(5)Exclude: since the values below are not between low and high

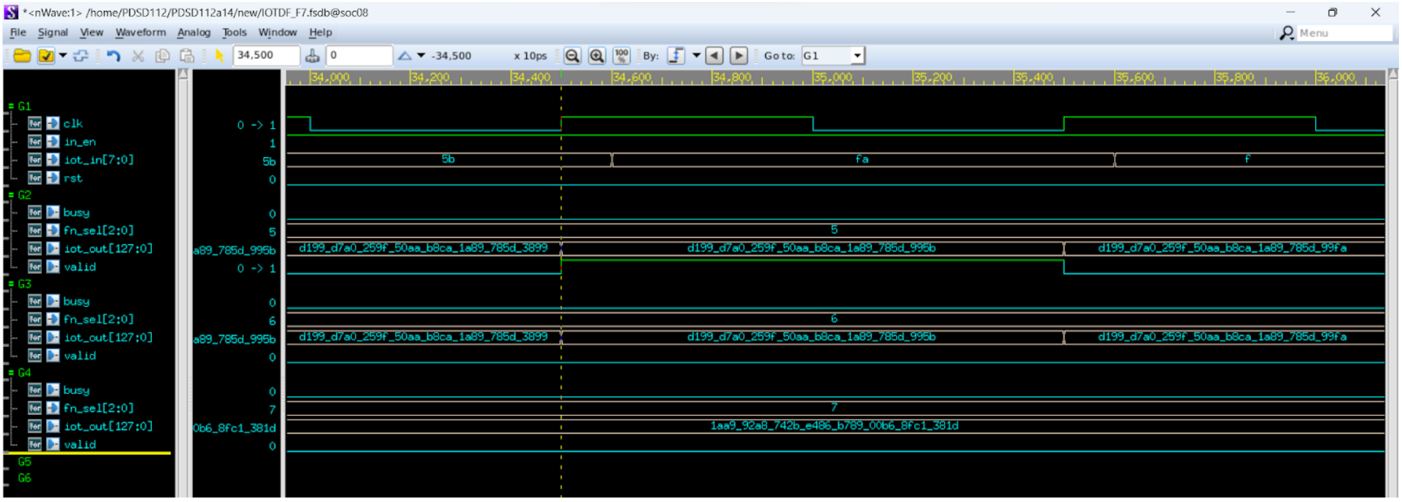
```
75_B3_11_8B_01_B9_3A_6D_6B_30_3C_41_B4_49_B7_2F      //Round_06
11_5A_5D_C0_3F_68_70_46_AA_2B_18_F3_BC_9F_8B_33      //Round_06
62_7D_FD_E0_9C_2F_19_FA_A2_AC_68_0B_A8_2E_BE_21      //Round_06
7F_C3_81_AE_9E_6B_41_B9_9D_79_59_84_57_D5_15_E4      //Round_06
D4_95_A1_68_F2_98_7F_A5_E0_66_73_E0_F6_7F_60_28      //Round_06
04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C      //Round_06
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93 .
All the patterns in round 6 aren't bigger than peakmax, therefore the PeakMax doesn't output anything.

=> correct!



(7)PeakMin:

Since the peakmin value now is 08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7 .
The patterns of 04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C in round 6 is smaller than peakmin, therefore the PeakMin output the following pattern

```
04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C      //Round_06
```

=> correct!

Round 7:

```
43_FD_AE_3D_3A_E5_05_87_C0_56_BC_8F_35_D7_AF_A1        //Round_07   P_00: 9.037546e+37
24_C4_21_03_91_2F_E0_D9_1D_70_34_99_27_1D_67_98        //Round_07   P_01: 4.887057e+37
08_96_11_61_8F_17_75_35_34_23_C4_13_CE_EF_8D_4B        //Round_07   P_02: 1.141302e+37
4A_CA_0B_E8_47_0E_FA_D3_1D_FA_3D_64_11_B0_9E_C8        //Round_07   P_03: 9.941196e+37
E4_8C_8D_4D_DF_0C_91_34_D7_70_9D_5A_7D_4E_6A_61        //Round_07   P_04: 3.037938e+38
29_13_E4_DE_D8_AE_DF_79_61_CD_58_3E_99_DC_D8_FC        //Round_07   P_05: 5.460164e+37
FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34        //Round_07   P_06: 3.369260e+38
21_D9_88_4F_B9_ED_1E_58_12_3B_1E_1C_B0_AE_D0_C7        //Round_07   P_07: 4.499402e+37
```

(1)MAX :since the max value is :

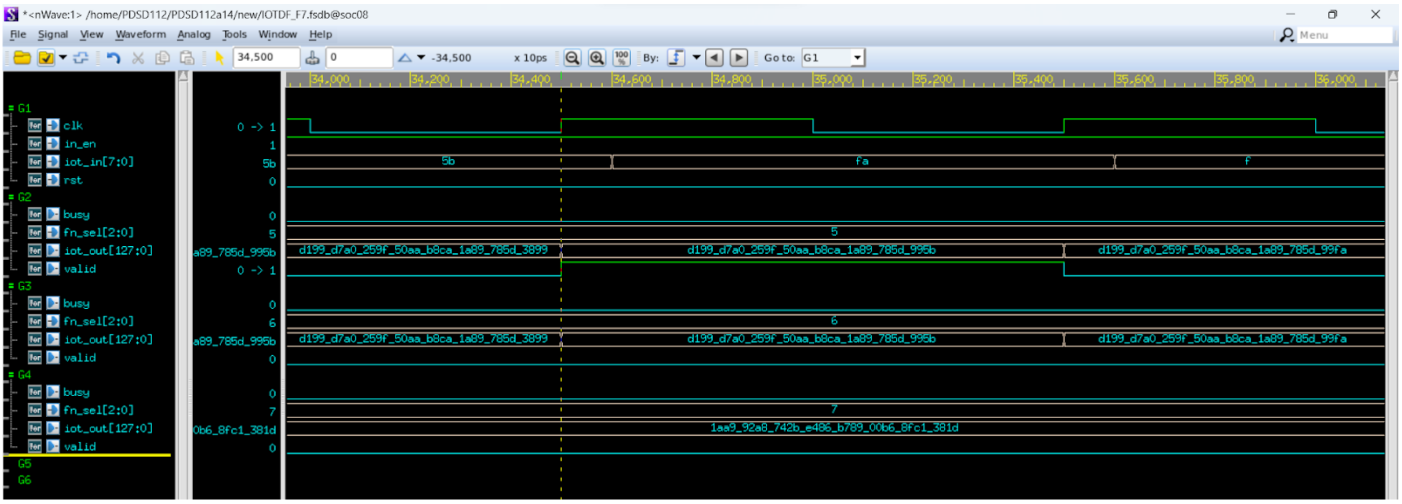
```
FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34        //Round_07
```

=> correct!

(2)MIN:since the min value is :

```
08_96_11_61_8F_17_75_35_34_23_C4_13_CE_EF_8D_4B        //Round_07
```

=> correct!



(3)AVE: since the average value is :

```
5D_22_AF_85_97_2B_A4_57_7E_05_F7_CE_2C_6C_3C_94_        //Round_07
```

=> correct!

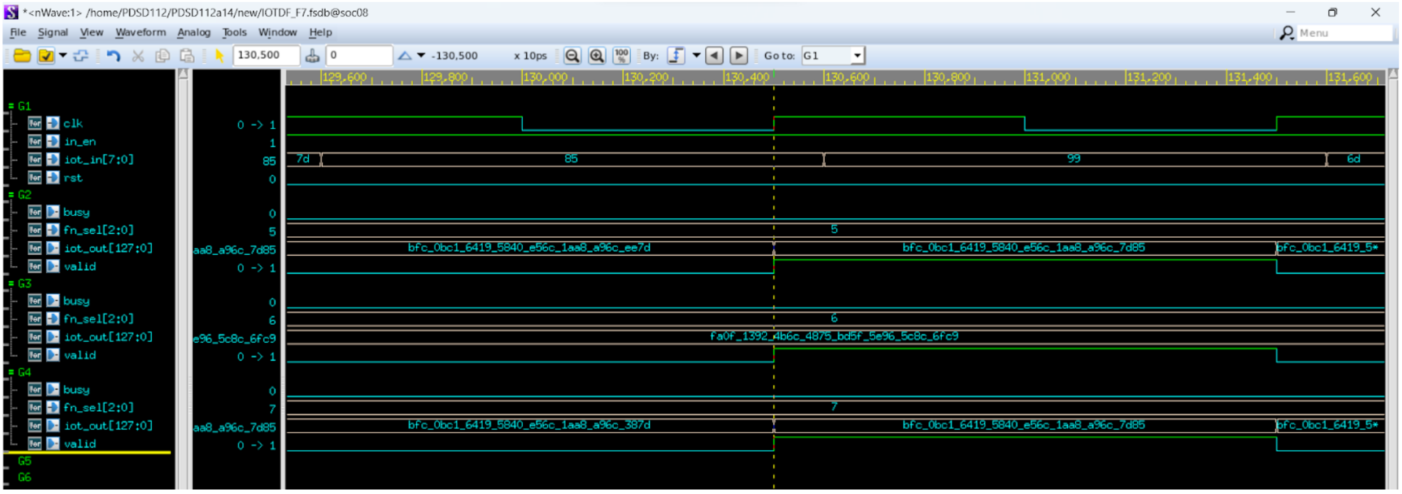
(4)Extract: since there is no pattern between low and high, output nothing

=> correct!

(5)Exclude: since the values below are not between low and high

```
43_FD_AE_3D_3A_E5_05_87_C0_56_BC_8F_35_D7_AF_A1      //Round_07
24_C4_21_03_91_2F_E0_D9_1D_70_34_99_27_1D_67_98      //Round_07
08_96_11_61_8F_17_75_35_34_23_C4_13_CE_EF_8D_4B      //Round_07
4A_CA_0B_E8_47_0E_FA_D3_1D_FA_3D_64_11_B0_9E_C8      //Round_07
E4_8C_8D_4D_DF_0C_91_34_D7_70_9D_5A_7D_4E_6A_61      //Round_07
29_13_E4_DE_D8_AE_DF_79_61_CD_58_3E_99_DC_D8_FC      //Round_07
FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34      //Round_07
21_D9_88_4F_B9_ED_1E_58_12_3B_1E_1C_B0_AE_D0_C7      //Round_07
```

=> correct!



(6)PeakMax:

Since the peakmax value now is FA_0F_13_92_4B_6C_48_75_BD_5F_5E_96_5C_8C_6F_C9 . The patterns of FC_EA_3F_27_9D_92_5D_C5_2E_C4_C7_EE_6C_90_5B_93 in round 3 is bigger than peakmax, therefore the PeakMax output the following pattern

```
FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34      //Round_07
```

=> correct!

(7)PeakMin:

Since the peakmin value now is08_1E_B6_4A_09_75_B8_B2_A8_E0_FB_D3_B1_EC_F8_A7 .All the patterns in round 7 aren't smaller than peakmin, therefore the PeakMin doesn't output anything.

=> correct!

Round 8:

```
77_67_A4_6B_44_69_26_ED_06_A4_2E_44_DB_90_50_D8      //Round_08  P_00: 1.587163e+38
5E_3C_68_35_1F_DE_0F_07_F6_41_31_44_E6_26_DB_76      //Round_08  P_01: 1.252611e+38
A8_8A_42_C4_26_B5_33_B8_28_1F_5C_EF_F3_FC_37_AD      //Round_08  P_02: 2.240282e+38
DF_5E_9B_3E_AA_F1_DE_60_EF_68_67_2F_E6_D0_1D_CC      //Round_08  P_03: 2.969091e+38
4B_C2_A1_E2_56_7A_C5_96_3C_11_39_81_01_E9_15_3C      //Round_08  P_04: 1.007027e+38
45_C4_C2_4C_9E_6F_EE_54_13_1C_AB_25_0A_AF_27_C0      //Round_08  P_05: 9.273836e+37
14_58_3B_30_C6_83_14_DD_AC_87_F5_EB_DE_4D_CD_EB      //Round_08  P_06: 2.704268e+37
2F_F7_49_B9_BB_86_11_DA_15_36_6E_ED_71_BB_F6_FE      //Round_08  P_07: 6.375771e+37
```

(1)MAX :since the max value is :

```
DF_5E_9B_3E_AA_F1_DE_60_EF_68_67_2F_E6_D0_1D_CC      //Round_08
```

=> correct!

(2)MIN:since the min value is :

```
14_58_3B_30_C6_83_14_DD_AC_87_F5_EB_DE_4D_CD_EB      //Round_08
```

=> correct!

(3)AVE: since the average value is :

```
66_6C_7A_77_95_9C_44_56_04_AB_2D_A5_1F_24_B0_75_      //Round_08
```

=> correct!

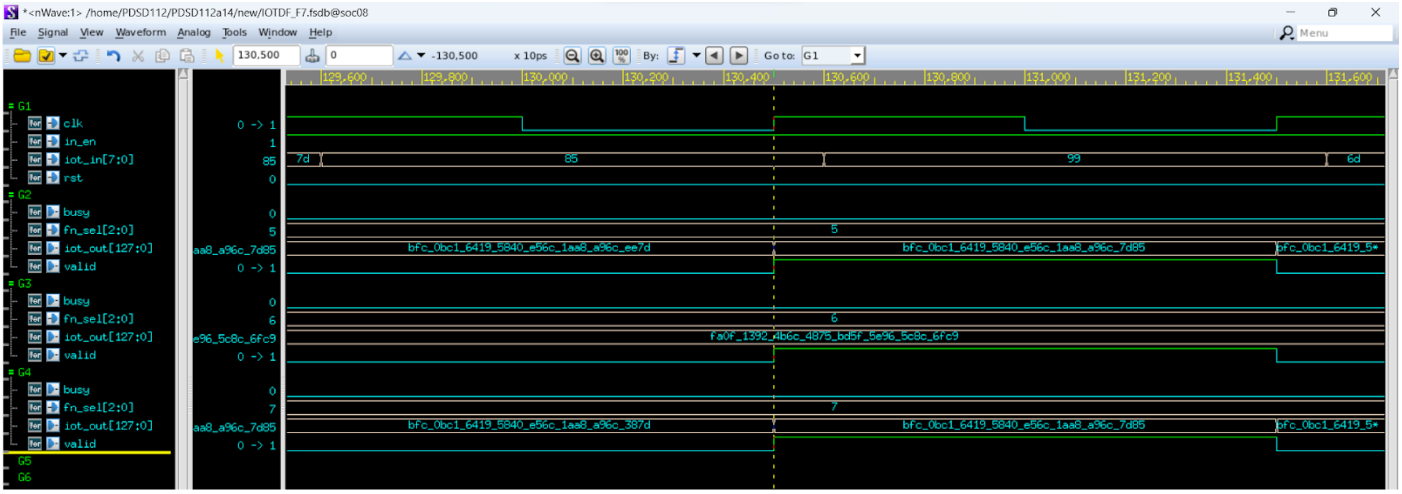
(4)Extract: since the between low and high show in the following

```
77_67_A4_6B_44_69_26_ED_06_A4_2E_44_DB_90_50_D8      //Round_08
A8_8A_42_C4_26_B5_33_B8_28_1F_5C_EF_F3_FC_37_AD      //Round_08
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
77_67_A4_6B_44_69_26_ED_06_A4_2E_44_DB_90_50_D8      //Round_08
5E_3C_68_35_1F_DE_0F_07_F6_41_31_44_E6_26_DB_76      //Round_08
DF_5E_9B_3E_AA_F1_DE_60_EF_68_67_2F_E6_D0_1D_CC      //Round_08
4B_C2_A1_E2_56_7A_C5_96_3C_11_39_81_01_E9_15_3C      //Round_08
45_C4_C2_4C_9E_6F_EE_54_13_1C_AB_25_0A_AF_27_C0      //Round_08
14_58_3B_30_C6_83_14_DD_AC_87_F5_EB_DE_4D_CD_EB      //Round_08
2F_F7_49_B9_BB_86_11_DA_15_36_6E_ED_71_BB_F6_FE      //Round_08
```



=> correct!

(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 .All the patterns in round 4 aren’t bigger than peakmax, therefore the PeakMax doesn’t output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C .All the patterns in round 4 aren’t smaller than peakmin, therefore the PeakMin doesn’t output anything.

=> correct!

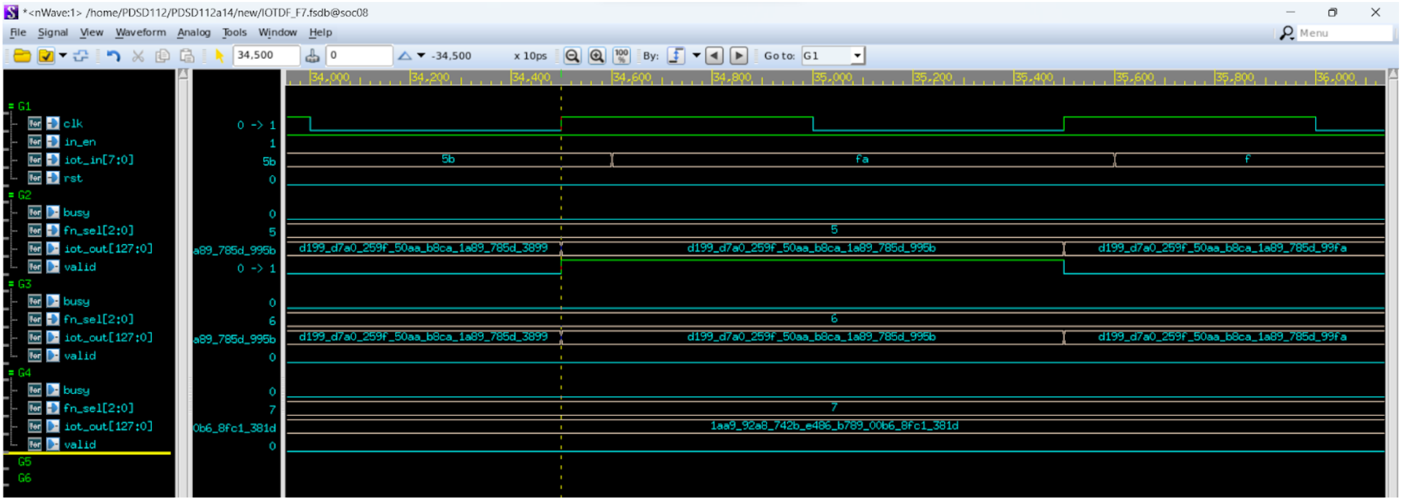
Round 9:

```
C2_5B_05_F4_BC_78_FA_8C_0F_06_CD_23_71_46_32_E2     //Round_09   P_00: 2.583429e+38
E8_31_4A_E6_0F_F5_85_0B_B1_FE_CC_DF_54_9D_C7_80     //Round_09   P_01: 3.086368e+38
A5_17_5A_B3_3A_DB_C9_88_00_BE_FE_03_F0_4B_06_01     //Round_09   P_02: 2.194439e+38
84_50_A6_D0_77_4E_4F_D3_28_FB_36_B8_B6_84_1D_03     //Round_09   P_03: 1.758769e+38
B3_F5_9B_DD_E6_B8_1E_90_46_F3_54_E0_F0_BD_2F_18     //Round_09   P_04: 2.392071e+38
0B_A3_C1_E3_8B_DA_7C_38_50_78_C3_63_87_BE_EC_C9     //Round_09   P_05: 1.547178e+37
E3_35_E2_3A_0E_4E_6A_52_B6_FC_C1_3F_05_67_8C_8D     //Round_09   P_06: 3.020145e+38
AC_55_E5_A0_CB_28_EA_10_92_DA_F6_1C_D2_29_07_B8     //Round_09   P_07: 2.290732e+38
```

(1)MAX :since the max value is :

```
E8_31_4A_E6_0F_F5_85_0B_B1_FE_CC_DF_54_9D_C7_80      //Round_09
```

=> correct!



(2)MIN:since the min value is :

```
0B_A3_C1_E3_8B_DA_7C_38_50_78_C3_63_87_BE_EC_C9      //Round_09
```

=> correct!

(3)AVE: since the average value is :

```
A4_63_2E_FF_59_54_51_03_D9_80_73_CB_F7_97_F9_B1_      //Round_09
```

=> correct!

(4)Extract: since the between low and high show in the following

```
A5_17_5A_B3_3A_DB_C9_88_00_BE_FE_03_F0_4B_06_01        //Round_09
84_50_A6_D0_77_4E_4F_D3_28_FB_36_B8_B6_84_1D_03        //Round_09
AC_55_E5_A0_CB_28_EA_10_92_DA_F6_1C_D2_29_07_B8        //Round_09
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
C2_5B_05_F4_BC_78_FA_8C_0F_06_CD_23_71_46_32_E2        //Round_09
E8_31_4A_E6_0F_F5_85_0B_B1_FE_CC_DF_54_9D_C7_80        //Round_09
0B_A3_C1_E3_8B_DA_7C_38_50_78_C3_63_87_BE_EC_C9        //Round_09
E3_35_E2_3A_0E_4E_6A_52_B6_FC_C1_3F_05_67_8C_8D        //Round_09
```

=> correct!

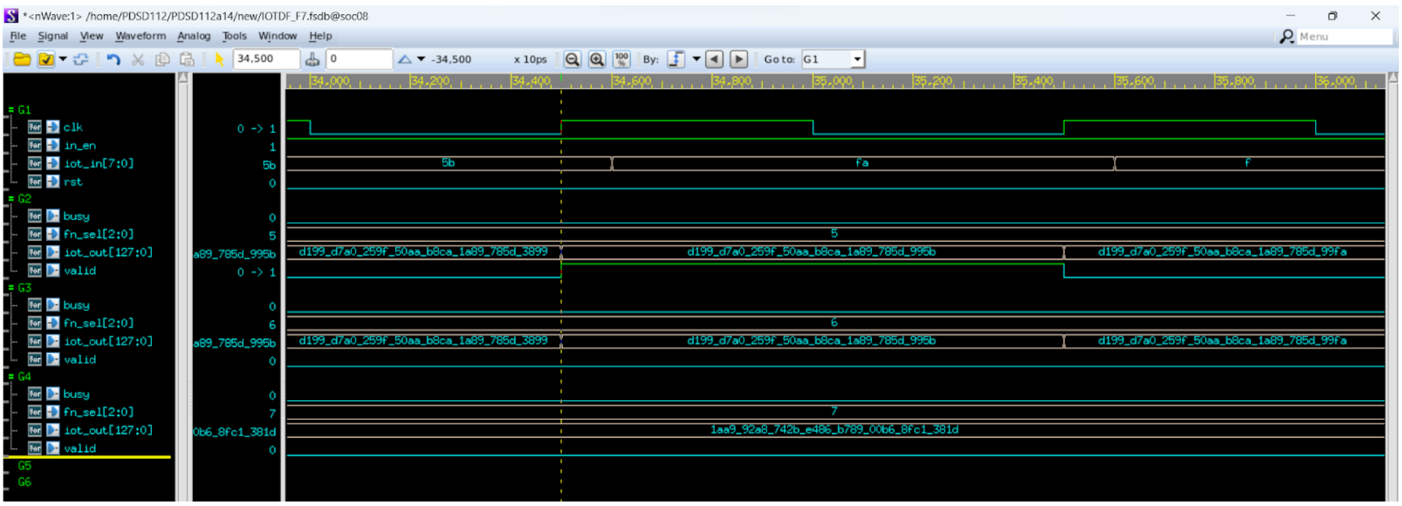
(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 .All the patterns in round 9 aren’t bigger than peakmax, therefore the PeakMax doesn’t output anything.

=> correct!

Round 10:

```
2A_57_13_3E_C6_13_28_5F_83_4F_91_D8_49_05_E0_3A     //Round_10   P_00: 5.627970e+37
80_E5_86_71_82_BF_34_42_93_F3_9B_A8_DE_1B_F3_03     //Round_10   P_01: 1.713329e+38
88_81_0B_C8_06_88_26_42_01_1D_5D_C7_DC_EE_7B_1E     //Round_10   P_02: 1.814451e+38
9F_8F_CB_B7_F9_74_F5_78_5A_1E_57_5B_8C_9D_6E_A2     //Round_10   P_03: 2.120939e+38
F6_CA_50_3E_3E_CA_BF_18_8B_00_99_2F_75_A2_9E_03     //Round_10   P_04: 3.280406e+38
91_59_3C_BC_E6_38_34_10_17_4D_86_02_D9_01_3A_F0     //Round_10   P_05: 1.932014e+38
93_38_44_E8_72_4E_AA_D1_45_5E_7C_15_59_37_B3_FF     //Round_10   P_06: 1.956887e+38
EF_D2_24_D7_10_BB_38_11_97_1E_A7_54_3D_14_FB_60     //Round_10   P_07: 3.187766e+38
```



(1)MAX :since the max value is :

```
F6_CA_50_3E_3E_CA_BF_18_8B_00_99_2F_75_A2_9E_03        //Round_10
```

=> correct!

(2)MIN:since the min value is :

```
2A_57_13_3E_C6_13_28_5F_83_4F_91_D8_49_05_E0_3A        //Round_10
```

=> correct!

(3)AVE: since the average value is :

```
9B_CF_6C_FD_5E_1B_89_CC_FE_29_44_A8_0E_B3_C8_A9_        //Round_10
```

=> correct!

(4)Extract: since the between low and high show in the following

```
80_E5_86_71_82_BF_34_42_93_F3_9B_A8_DE_1B_F3_03        //Round_10
88_81_0B_C8_06_88_26_42_01_1D_5D_C7_DC_EE_7B_1E        //Round_10
9F_8F_CB_B7_F9_74_F5_78_5A_1E_57_5B_8C_9D_6E_A2        //Round_10
91_59_3C_BC_E6_38_34_10_17_4D_86_02_D9_01_3A_F0        //Round_10
93_38_44_E8_72_4E_AA_D1_45_5E_7C_15_59_37_B3_FF        //Round_10
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
2A_57_13_3E_C6_13_28_5F_83_4F_91_D8_49_05_E0_3A        //Round_10
F6_CA_50_3E_3E_CA_BF_18_8B_00_99_2F_75_A2_9E_03        //Round_10
EF_D2_24_D7_10_BB_38_11_97_1E_A7_54_3D_14_FB_60        //Round_10
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 .All the patterns in round 10 aren’t bigger than peakmax, therefore the PeakMax doesn’t output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C .All the patterns in round 10 aren't smaller than peakmin, therefore the PeakMin doesn't output anything.

=> correct!

Round 11:

```
0F_13_31_F8_19_75_73_4C_01_B6_09_4D_0D_29_29_E5     //Round_11  P_00: 2.003809e+37
A8_15_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97     //Round_11  P_01: 2.234237e+38
19_4F_BE_DB_83_40_94_74_60_52_ED_62_94_22_A4_D5     //Round_11  P_02: 3.364476e+37
9B_DB_BE_C3_61_94_25_E9_2A_C9_95_EF_E7_12_A2_7B     //Round_11  P_03: 2.071713e+38
CD_79_44_C2_00_CE_A1_8C_4E_AB_03_28_C5_44_FC_0F     //Round_11  P_04: 2.731214e+38
08_53_1D_75_B7_0C_2F_6D_45_1C_4B_C7_B4_0A_63_75     //Round_11  P_05: 1.106538e+37
85_75_0C_62_58_26_15_66_4E_B9_0B_2D_6C_BE_43_51     //Round_11  P_06: 1.773951e+38
4C_AC_46_1B_A8_7D_36_10_95_E1_CB_CE_F5_F1_75_E9     //Round_11  P_07: 1.019158e+38
```

(1)MAX :since the max value is :

```
CD_79_44_C2_00_CE_A1_8C_4E_AB_03_28_C5_44_FC_0F      //Round_11
```

=> correct!

(2)MIN:since the min value is :

```
08_53_1D_75_B7_0C_2F_6D_45_1C_4B_C7_B4_0A_63_75      //Round_11
```

=> correct!

(3)AVE: since the average value is :

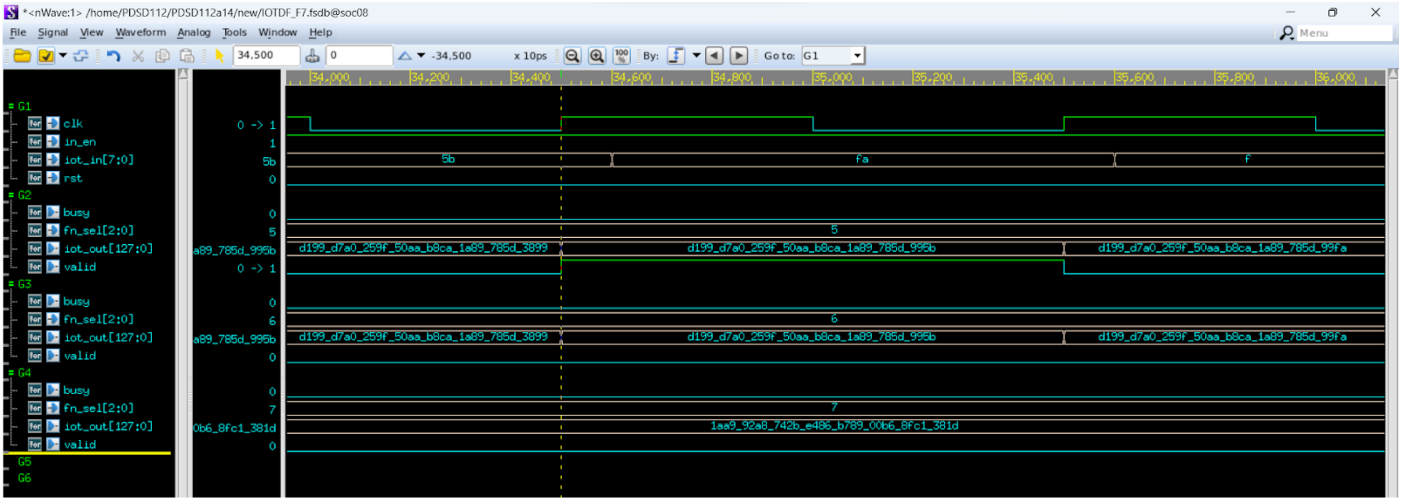
```
62_88_47_2E_22_A6_07_5C_61_A5_3C_61_B0_CE_A5_71_      //Round_11
```

=> correct!

(4)Extract: since the between low and high show in the following

```
A8_15_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97        //Round_11
9B_DB_BE_C3_61_94_25_E9_2A_C9_95_EF_E7_12_A2_7B        //Round_11
85_75_0C_62_58_26_15_66_4E_B9_0B_2D_6C_BE_43_51        //Round_11
```

=> correct!



(5)Exclude: since the values below are not between low and high

```
0F_13_31_F8_19_75_73_4C_01_B6_09_4D_0D_29_29_E5        //Round_11
19_4F_BE_DB_83_40_94_74_60_52_ED_62_94_22_A4_D5        //Round_11
CD_79_44_C2_00_CE_A1_8C_4E_AB_03_28_C5_44_FC_0F        //Round_11
08_53_1D_75_B7_0C_2F_6D_45_1C_4B_C7_B4_0A_63_75        //Round_11
4C_AC_46_1B_A8_7D_36_10_95_E1_CB_CE_F5_F1_75_E9        //Round_11
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 .All the patterns in round 11 aren't bigger than peakmax, therefore the PeakMax doesn't output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C .All the patterns in round 11 aren't smaller than peakmin, therefore the PeakMin doesn't output anything.

=> correct!

2. Corner Case Considerations and Automatically Checking Testbench:

Round 12:

```
0F_13_31_F8_19_75_73_4C_01_B6_09_4D_0D_29_29_E5      //Round_12 P_00: 2.003809e+37
0F_12_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97      //Round_12 P_01: 2.234237e+38
0F_15_BE_DB_83_40_94_74_60_52_ED_62_94_22_A4_D5      //Round_12 P_02: 3.364476e+37
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_03: 2.071713e+38
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_04: 2.731214e+38
6F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_05: 1.106538e+37
7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_06: 1.773951e+38
7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_07: 1.019158e+38
```

(1)MAX :since the max value is :

```
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12
```

=> correct!

(2)MIN:since the min value is :

```
0F_12_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97      //Round_12
```

=> correct!

(3)AVE: since the average value is :

```
0F_12_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97      //Round_12
```

=> correct!

(4)Extract: since the between low and high show in the following

```
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1       //Round_12
7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_12
7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1       //Round_12
```

=> correct!

(5)Exclude: since the values below are not between low and high

```
0F_13_31_F8_19_75_73_4C_01_B6_09_4D_0D_29_29_E5      //Round_12 P_00: 2.003809e+37
0F_12_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97      //Round_12 P_01: 2.234237e+38
0F_15_BE_DB_83_40_94_74_60_52_ED_62_94_22_A4_D5      //Round_12 P_02: 3.364476e+37
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_03: 2.071713e+38
AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_04: 2.731214e+38
6F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_05: 1.106538e+37
7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_07: 1.019158e+38
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 .All the patterns in round 12 aren’t bigger than peakmax, therefore the PeakMax doesn’t output anything.

=> correct!

(7)PeakMin:

Since the peakmin value now is04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C .All the patterns in round 12 aren’t smaller than peakmin, therefore the PeakMin doesn’t output anything.

=> correct!

Round 13:

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_00: 2.003809e+37
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_01: 2.234237e+38
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_02: 3.364476e+37
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_03: 2.071713e+38
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_04: 2.731214e+38
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_05: 1.106538e+37
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_06: 1.773951e+38
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_07: 1.019158e+38
```

(1)MAX :since the max value is :

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
```

=> correct!

(2)MIN:since the min value is :

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
```

=> correct!

(3)AVE: since the average value is :

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
```

=> correct!

(4)Extract: since there is no pattern between low and high, output nothing

=> correct!

(5)Exclude: since the values below are not between low and high

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13
```

=> correct!

(6)PeakMax:

Since the peakmax value now is FD_79_95_25_A5_79_3D_4C_74_D1_B8_1B_5D_F2_8D_34 . The patterns of FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF in round 13 is bigger than peakmax, therefore the PeakMax output the following pattern

```
FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF       //Round_13   P_00: 2.003809e+37
```

=> correct!

(7)PeakMin:

Since the peakmin value now is04_89_64_0C_5F_83_85_15_C4_30_33_66_1F_F6_67_6C .All the patterns in round 15 aren't smaller than peakmin, therefore the PeakMin doesn't output anything.

=> correct!

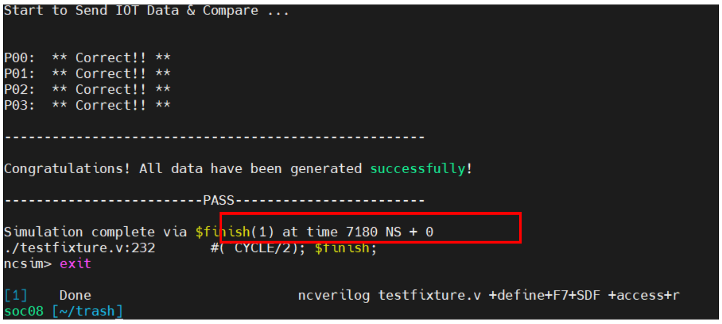
Analyze:

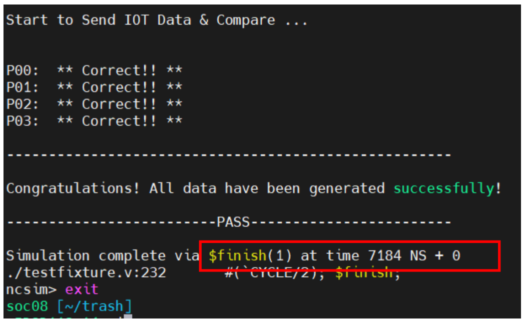
```
 97  0F_13_31_F8_19_75_73_4C_01_B6_09_4D_0D_29_29_E5      //Round_12 P_00: 2.003809e+37
 98  0F_12_D5_24_5E_67_F0_C9_07_F5_30_81_22_17_A1_97      //Round_12 P_01: 2.234237e+38
 99  0F_15_BE_DB_83_40_94_74_60_52_ED_62_94_22_A4_D5      //Round_12 P_02: 3.364476e+37
100  AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_03: 2.071713e+38
101  AF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_04: 2.731214e+38
102  6F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_05: 1.106538e+37
103  7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_12 P_06: 1.773951e+38
104  7F_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_F1      //Round_12 P_07: 1.019158e+38
105  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_00: 2.003809e+37
106  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_01: 2.234237e+38
107  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_02: 3.364476e+37
108  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_03: 2.071713e+38
109  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_04: 2.731214e+38
110  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_05: 1.106538e+37
111  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_06: 1.773951e+38
112  FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF_FF      //Round_13  P_07: 1.019158e+38
```

We add corner case test. Like line 100 and 101. Since (F4) extract upper bound is AF followed by all F. Line 100 will become output while line 101 doesn't. And round 13 are all-F patterns, which can test the maximum input of F3(average). Line 97~99 shows that the 0F is the same, but followed by different value, so they're also a corner case of F1(max) and F2(min), since the change decision can't just be made by the MSB input, if the input is equal to the previous one, we should declare it as undefined and observe by the following inputs.

```
`define SDFFILE     "/hom
`define CYCLE       4
`define DEL         1.0
`define PAT_NUM     112
`define F1_NUM      14
`define F4_NUM      31
`define F5_NUM      88
`define F6_NUM      4
`define F7_NUM      4
```

We only need to adjust the pattern number of the correct answer in fn.dat. If we don't modify the length of FN_NUM, the testbench would stuck and we have to exit the program. However, if we modify the length of FN_NUM, the testbench can check the answer automatically. Furthermore, test patterns can be generated by other languages, such as python or c++, which we're more familiar with, it's also the advantages of testbench, allowing us to read files from outside.





As we can see, the time the test done is different from the result in part 5. It's because we modified the test cycle, so the simulation time becomes longer, however, it'll check the answer automatically.

IV. Logic Synthesis Process and Synthesis Parameters

This time, we utilize tcl file to compile automatically.

```
1  set company "CIC"
2  set designer "Student"
3  set search_path "/mnt3/CBDK_IC_Contest_v2.1/SynopsysDC/db $search_path"
4  set link_library "slow.db fast.db dw_foundation.sldb"
5  set target_library "slow.db fast.db"
6  set symbol_library "generic.sdb"
7  set synthetic_library "dw_foundation.sldb"
8
9  set hdlin_translate_off_skip_text "TRUE"
10 set edifout_netlist_only "TRUE"
11 set verilogout_no_tri true
12
13 set hdlin_enable_presto_for_vhdl "TRUE"
14 set sh_enable_line_editing true
15 set sh_line_editing_mode emacs
16 history keep 100
17 alias h history
18
19 set bus_inference_style {%s[%d]}
20 set bus_naming_style {%s[%d]}
21 set hdlout_internal_busses true
22 define_name_rules name_rule -allowed {a-z A-Z 0-9 _} -max_length 255 -type cell
23 define_name_rules name_rule -allowed {a-z A-Z 0-9 _[]} -max_length 255 -type net
24 define_name_rules name_rule -map {{"\\*cell\\*" "cell"}}
25
```

For line 1 to 25, it's the dc setup from the assignments before.

```
26 #Path_Top:Verilog放置的位置
27 #Path_Syn:合成後report.txt檔案要放置的根位置，需自行在目錄下創建名為dc_out_file之資料夾
28 #Dump_file_name:合成後產生檔案之名字
29 set Path_Top        "/home/PDSD112/PDSD112a14/trash/"
30 set Path_Syn        "/home/PDSD112/PDSD112a14/trash/syn/"
31 set Dump_file_name  "IOTDF_syn"
32 #設定Top module 名稱，需跟自行設計之電路的top module name相同
33 set Top             "IOTDF"
34 set optimize        "_2"
35 #Specify Clock，clock名需和top module中clk port相同
36 set Clk_pin         "clk"
37 set Clk_period      "4"
38
```

Path_Top: the path where the top module stored.
Path_Syn: the path where the synthesized module stored.
Dump_file_name: the name where the synthesized module.
Top: same name as our tom module
Optimize: I add it by myself, which can let me know the different result, and the file won't be covered

Clk_pin: name of the clock pin
Clk_period: set the clock period

```
39  #Read Design
40  #如果設計有parameter設計，read_file指定不能用，需使用analyze + elaborate指令並自行更改路徑
41  # read_file -format verilog {/home/m103040049/HDL_HW/multiplier.v}
42  # current_design $Top
43  analyze -format verilog {"/home/PDSD112/PDSD112a14/trash/IOTDF.v"}
44  elaborate $Top
45
46  #檢查是否讀取成功
47  link
48  set_fix_multiple_port_nets -all -buffer_constants
49  #Max Delay (For Combinational Circurt)
50  # set_max_delay $Clk_period  -from [all_inputs] -to [all_outputs]
51  # create_clock -name $Clk_pin -period $Clk_period
52
53  #Setting Timing Constraints、Specify Clock (For Sequential Circurt)
54   create_clock -name $Clk_pin -period $Clk_period [get_ports $Clk_pin]
55   set_dont_touch_network                        [get_clocks $Clk_pin]
56   set_fix_hold                                      [get_clocks $Clk_pin]
57   set_ideal_network                                 [get_ports $Clk_pin]
58
59  #Setting Input / Output Delay
60  set_input_delay     0    -clock $Clk_pin [remove_from_collection [all_inputs] [get_ports $Clk_pin]]
61  set_output_delay    0    -clock $Clk_pin [all_outputs]
```

Line 43 analyze -format verilog {"all files required"}
elaborate $Top
Line 48 separate wire with same name of input and output by buffer
Line53~57: set the clock attribute
Line60~61: set the input and output delay

```
62
63  #Setting Operating Conditions
64  set_operating_conditions -min_library slow -min slow  -max_library slow -max slow
65
66  #Setting Wire Load Model
67  set_wire_load_model -name tsmc13_wl10 -library slow
68  set_wire_load_mode top
69
70  #若合成時只考慮面積而不在意時間，則可以不設定Timing Constraints，只設定Max area為0
71  #set_max_area 0
72
73  compile
74
75  #Change Naming Rule
76  set bus_inference_style {%s[%d]}
77  set bus_naming_style {%s[%d]}
78  set hdlout_internal_busses true
79  change_names -hierarchy -rule verilog
80  define_name_rules name_rule -allowed "A-Z a-z 0-9 _" -max_length 255 -type cell
81  define_name_rules name_rule -allowed "A-Z a-z 0-9 _[]" -max_length 255 -type net
82  define_name_rules name_rule -map {{"\\*cell\\*""cell"}}
83  define_name_rules name_rule -case_insensitive
84  change_names -hierarchy -rules name_rule
85  remove_unconnected_ports -blast_buses [get_cells -hierarchical *]
86
```

Line64: set the operating condition
Line 67~68: set the wire load
Line73: compile
Line 75~85 : change the format

```
87  #Report
88  report_timing -path full -delay max -max_paths 1 -nworst 1 > $Path_Syn/timing_report_${Dump_file_name}${optimize}.txt
89  report_area -hier -nosplit > $Path_Syn/area_report_${Dump_file_name}${optimize}.txt
90  report_power -analysis_effort low > $Path_Syn/power_report_${Dump_file_name}${optimize}.txt
91
92  #Write out
93  write -hierarchy -format ddc -output $Path_Syn/${Dump_file_name}${optimize}.ddc
94  write -format verilog -hierarchy -output $Path_Syn/${Dump_file_name}${optimize}.v
95  write_sdf -version 2.1 -context verilog $Path_Syn/${Dump_file_name}${optimize}.sdf
96  write_sdc $Path_Syn/${Dump_file_name}${optimize}.sdc
```

Line 88~96: set the output file, including reports and synthesized files

```
soc08 [~]
-PDSD112a14- $cd /home/PDSD112/PDSD112a14/trash/
soc08 [~/trash]
-PDSD112a14- $dc_shell -f dc.tcl
```

Cd to the folder and key: dc_shell -f dc.tcl , then the compile starts.

V. Logic Synthesis and Verification Results

1. Area

```
****************************************
Report : area
Design : IOTDF
Version: P-2019.03
Date   : Mon Jan  8 05:14:56 2024
****************************************

Library(s) Used:

    slow (File: /mnt3/CBDK_IC_Contest_v2.1/SynopsysDC/db/slow.db)

Number of ports:                         9105
Number of nets:                         12765
Number of cells:                         4359
Number of combinational cells:           3262
Number of sequential cells:               399
Number of macros/black boxes:               0
Number of buf/inv:                       1237
Number of references:                     142

Combinational area:              30716.150323
Buf/Inv area:                     8739.912493
Noncombinational area:           13847.389460
Macro/Black Box area:                0.000000
Net Interconnect area:          358495.160980

Total cell area:                 44563.539783
Total area:                     403058.700763
```

2. Simulation Time

<table>
<tr>
<td>F1</td>
<td>
<pre>
Congratulations! All data have been generated successfully!

-----------------------PASS-----------------------

Simulation complete via $finish(1) at time 6160 NS + 0
./testfixture.v:232   #(CYCLE/2); $finish;
ncsim> exit

[1]   Done   ncverilog testfixture.v +define+F1+SDF +access+r
soc08 [~/trash]
</pre>
</td>
</tr>
</table>

F2	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6160 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[1]    Done       ncverilog testfixture.v +define+F2+SDF +access+r``
	``soc08 [~/trash]``

F3	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6164 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[1]    Done       ncverilog testfixture.v +define+F3+SDF +access+r``
	``soc08 [~/trash]``

F4	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6156 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[2]    Done       ncverilog testfixture.v +define+F4+SDF +access+r``
	``soc08 [~/trash]``

F5	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6160 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[3]    Done       ncverilog testfixture.v +define+F5+SDF +access+r``
	``soc08 [~/trash]``

F6	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6156 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[4]    Done       ncverilog testfixture.v +define+F6+SDF +access+r``
	``soc08 [~/trash]``

F7	``Congratulations! All data have been generated successfully!``
	``------------------------PASS------------------------``
	``Simulation complete via $finish(1) at time 6156 NS + 0``
	``./testfixture.v:232       #(`CYCLE/2); $finish;``
	``ncsim> exit``
	``[5]    Done       ncverilog testfixture.v +define+F7+SDF +access+r``
	``soc08 [~/trash]``

3. Area x Simulation Time

Area: 44563.539783 um^2

Function 1 Simulation Time: 6160ns

Function 2 Simulation Time: 6160ns
Function 3 Simulation Time: 6164ns
Function 4 Simulation Time: 6156ns
Function 5 Simulation Time: 6160ns
Function 6 Simulation Time: 6156ns
Function 7 Simulation Time: 6156ns

Area x (Sum of Simulation Time) = 1,921,223,327.124696

VI. Reflection and Discussion

(Each team member at least 500 words)

[Team Member 1]洪漢霖

In this project, the most important thing that I learned was the value of collaboration with teammates. Although the project itself was not particularly challenging compared to previous projects, we faced several obstacles that tested our teamwork and time management skills. As a team, we had to balance our workload with other commitments such as exams and other projects, which made it difficult to dedicate enough time to perfect our design.

At the beginning of the project, my teammate Yeh and I tried our best to write hardware-oriented code, but unfortunately, we encountered some bugs that set us back and we can't fix it before the deadline. We were both disappointed with ourselves, but we learned from the experience and adapted our approach to a software-style coding. Our teammate Lai, on the other hand, continued to work diligently throughout the project, and her dedication was a source of inspiration for the rest of us.

Despite the challenges we faced, we were able to deliver a concept presentation that was well-received by our instructor. Our dream was to become the best group in the class, but we fell short of our goal due to time constraints and other commitments. However, I am grateful for the opportunity to work with Yeh in the midnights, and I appreciate his dedication to the project.

We've designed 3 kinds of adder, since we think the average function is critical. We learned 3 structures from the reference papers, some has lower area, some have lower delay. After comparing the influence of area and timing, we utilize the adder which has the larger area but lower delay. Dramatically, comparing with other teams, we find that the adder isn't having big difference between the "C=A+B" one and the specific structure, which is quite frustrating!

Through this project, I learned that time management is crucial to success in any endeavor. We made our report just before the deadline, which put us under a lot of pressure. I also learned that failure is an essential part of the learning process. Although we did not achieve our desired outcome, we gained valuable experience and knowledge that will help us in our future endeavors.

In conclusion, I regard this project as a wonderful journey that allowed me to collaborate with my teammates. The experience taught me that I still have a lot to learn to become a successful IC designer, and I am determined to put in the effort required to achieve my goals. I am grateful for the opportunity to work with my teammates and for the lessons we learned together. In the future work, we must have the capability to co-work with each other and make good management of our projects, especially the design time. Besides, the specification of the project is crucial, too. We must try our best to meet the specification.
The feedback from the other teams is also helpful to us, some let us know where we didn't give a clear explanation, some praise us with the vivid slides and the animation.

[Team Member 2] 葉米亞

The final project for this semester proved to be a formidable challenge. In the initial stages, Hong and I had smooth discussions regarding the circuit architecture, fostering optimism about achieving optimal performance. However, as we delved into the implementation of actual code, we encountered difficulties translating our conceptualized ideas into RTL code. Various minor issues surfaced, and identifying them wasn't always straightforward, underscoring the critical importance of meticulous debugging.

Even seemingly minor mistakes, such as errors in the "always" conditions, had the potential to cascade into broader issues throughout the entire circuit. Uncovering such subtle problems demanded a considerable amount of time, occasionally even compromising our ability to get adequate rest. Nevertheless, this process equipped us with valuable debugging skills, particularly in utilizing nWave to observe signal changes and propagation at each clock cycle to identify potential errors.

Within the scope of this project, my responsibility centered around implementing the most efficient adder. I developed three distinct algorithms for adders: the Kogge-Stone Adder, Han-Carlson Adder, and Brent-Kung Adder. Despite the Kogge-Stone Adder exhibiting an area larger by approximately 41% compared to the Han-Carlson Adder and Brent-Kung Adder, its delay was 53% less. Given that the overall score is determined by multiplying the area and delay, we concluded that employing the Kogge-Stone Adder would be the optimal choice.

This challenging experience taught us the importance of attention to detail, perseverance in debugging, and the value of choosing the right algorithm for a specific application, ultimately contributing to our growth as engineers.

The benefits of this semester's final project extend to various aspects of my future. Through collaboration with team members, I learned how to communicate and collaborate effectively in a team environment. The project process of solving difficulties together with teammates and reaching consensus cultivated my teamwork skills, providing valuable experience for future collaborative projects in the workplace.

Additionally, the continuous debugging process enhanced my debugging skills. Dealing with various issues, from syntax errors to logic errors, required careful analysis and resolution for each problem. This experience improved the accuracy of my Verilog code writing and deepened my understanding of the interaction between various components in a system.

Lastly, by implementing Verilog code for different adders and conducting synthesis, I gained a profound understanding of the hardware design process and principles. This knowledge helps me better apply hardware description languages (HDL) and engage in circuit design. I also learned how to balance circuit area and delay for optimal performance. These skills are valuable for future endeavors in hardware-related fields such as integrated circuit design or embedded system development.

[Team Member 3]賴佳莘

In this semester's courses, I have learned a lot—from the basics of structuring code to running simulations and waveforms on workstations, and ultimately mastering the synthesis techniques to achieve the desired performance for electronic circuits. Despite it being my second time taking this course, I still gained valuable insights. Concepts that were unclear to me in the previous semester became clear through the teacher's explanations and discussions with classmates. My favorite part was the class discussions, as they transformed the learning experience from mere listening to an interactive exploration of the topics. Through these discussions, I not only clarified areas of confusion but also discovered potential contradictions in my own viewpoints. After completing this course, I have gained a deeper understanding of IC design, and I believe it will greatly benefit me in my future endeavors.